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BEFORE YOU BEGIN THE SOFTWARE LOAD - CAUTIONS AND WARNINGS



NEW FOR 10.0 SOFTWARE! A KEY MUST BE INSTALLED BY THE FE BEFORE THE SYSTEM WILL PERMIT THE CUSTOMER TO SCAN. THE CUSTOMER WILL NOT BE ABLE TO RUN PATIENT SCANS ON A SYSTEM EQUIPPED WITH M4 SOFTWARE WITHOUT THIS KEY. SEE SECTION 11 FOR MORE DETAILS. DO NOT TURN THE SYSTEM OVER TO THE CUSTOMER UNLESS THIS KEY HAS FIRST BEEN INSTALLED.

Important: If your system is connected to a Hospital Network, disconnect it before proceeding. Failure to disconnect the SGI computer can corrupt the software load process.

Important: During the Install GUI phase of the software load, be sure the **<Num Lock>** key on the key board is turned off. If the **<Num Lock>** is on, the symptom seen is that nothing will happen when trying to type in any of the edit fields in the Install GUI.

Important: When reconfiguring the system using the Install/Configuration GUI, remember to push the **<Configure>** button on the verify tab before going to the Save info tab and doing a SaveInfo. If you have not done the final configuration by pushing the **<Configure>** button in the Verification tab, the information saved during Save Info will not reflect the latest changes made in the Install GUI.

Important: If at any time during the OS load a problem occurs that causes you to stop the install, correct the problem then reset the computer or power cycle the computer and begin the OS install again from the start. This is necessary to insure a complete load. If this does not fix your problem it may be necessary to do a low level format on the system drives. Refer to Appendix C for this process.

Important: If the site will be using the "YP" networking option, **YP must be turned off**, until after the InSite IIP option has been configured and checked out, or it will cause the InSite checkout to fail.

Important: If at the end of OS load the system will not boot and stops with the error, "/var/root partition corruption". Do the following:

1. Open a shell window.
2. Type at the prompt: **dd if=/dev/zero of=/dev/rdisk/dks0d1s0 count=1000 <Enter>**.
This writes zeros across the disk and will run for less than one minute.
3. Start the OS load from the beginning.

This error is a very rare event.

Important: Release 10.x or greater can only be loaded on Octane Computers.

1- LOADING HOST COMPUTER - OVERVIEW & PREPARATION

Note

If this is a new install or system hardware upgrade of a Signa system, the system software has been pre-loaded and will not need to be reloaded. Skip to section 4 for information on how to reconfigure the system for new installs or upgrades.

1-1 Required Media

- IRIX 6.5 Operating System CDROM Set, Disk 1 of 2 and Disk 2 of 2
- IRIX 6.5.4 Boot/Update CDROM
- MrApps - MR Applications CDROM
- MrpResSrv - MR Restricted Service CDROM (*GE Service use only*)
or MrpAdvSrv - MR Advanced Service CDROM (*Licensed In-House Customers*)
- saveINFO - MaxOptix MOD Media (Either blank or previously used)

DO NOT PROCEED UNLESS YOU HAVE ALL THIS MEDIA AVAILABLE

1-2 Archive Images

This installation instruction will erase any current images left in the database. **Archive all images BEFORE continuing with this procedure.**

1-3 Required Information Prior To Loading Software:

Vital site information is recorded to ensure proper software installation and configuration of the new software.

1. With the system up and running enter Guided Install.
2. Step through each tab. Record the values from each field using the Datasheet (Table 1-1).
3. The following parameters are determined by physical observation. Record results in Table 2-1.
 - **Package Name** (Echospeed SR150, Echospeed SR120, Highspeed SR77, Base SR20)
 - **Modem Type**
 - **Camera Type**
 - **DASM Type**
 - **Coil Type** (BRM or CRM)
4. If this is a Mobile System, select **[Mobile]**, then record all **Site Names** and **IP addresses**.

TABLE 1-1
SITE CONFIGURATION INFORMATION

TAB	ENTRY (✓ - Vital Info)	VALUE
Network Info	✓ (Top of Network tab)	__ Network Computer __ Standalone Computer
	✓ Host IP Address	
	✓ Netmask	(Default is 255.255.252.0)
	✓ Hostname	
	✓ Domain Name	
	YP(NIS) Service	__ On __ Off
	✓ IWS PC IP Address	
	Magnet Monitor IP Address	_____ (This parameter not used for anything but OpenSpeed systems. However, you must put in a placeholder value: (e.g. 0.0.0.0))
	✓ FE Laptop IP Address	
	✓ TPS/MGD Subnet (IP)	The TPS will only support IP addresses of: 10.0.XXX 10.47.XXX 10.121.XXX 10.217.XXX 172.22.XXX 192.168.XXX (Where XXX are numeric values chosen locally. The installer will need to verify the IP address used will not conflict the local sites network.)
	✓ Default Router IP Address	
	Power Monitor PC IP Address	_____ (This parameter not used for anything but 3T systems. However, you must put in a placeholder value: (e.g. 0.0.0.0))
System Configure	✓ Hospital Name	
	✓ Suite ID	
	✓ Host ID	
	✓ Unique ID	
	✓ Service ID	
	Date Format	
	Weight Unit	__ Pounds __ Kilograms
	Monitor Type	__ Gray __ Color
	Palette	__ Gray __ Color
	Keyboard	__ Amer. __ Eng. __ Fr. __ Ger. __ Ital. __ Port. __ Span. __ Swed.
	Language	__ Eng. __ Fr. __ Ger. __ Ital. __ Port. __ Span.
Hardware	✓ Field Strength	__ 0.35T __ 0.5T __ 0.7T __ 1.0T

TAB	ENTRY (✓ - Vital Info)	VALUE
Configure		☐ 1.5T ☐ 3.0T ☐ 4.0T
	✓ Package Name	☐ ECHOSPEED SR120 ☐ ECHOSPEED SR150 ☐ ECHOSPEED SR190 ☐ Base(HORIZON) SR20 ☐ Base(HORIZON) SR25 ☐ Base(HORIZON) SR40 ☐ SMARTSPEED SR50 ☐ HIGHSPEED SR77 ☐ TWINSPEED SR80/SR150
	✓ Gradient Type	☐ 8645 (Single or Double Bay) ☐ 8280 (SGD Base System) ☐ 8290 ☐ 8651 (SGD HighSlew) ☐ 8915 (ACGD) ☐ 8905 (ACGD LITE)
	✓ Resonance Module	☐ BRM ☐ CRM ☐ ORM ☐ SPRM ☐ TRM ☐ MFO ☐ miniCRM
	Magnet Serial Number	
	Table Limit	
	Line Frequency	☐ 60Hz ☐ 50Hz
	✓ RF Amp Type	☐ 1.0T Teli Solid State RF SRFD ☐ 1.5T Analogic Solid State RF SRFD ☐ 3.0T AMT RFAMP ☐ 0.7T Analogic RFAMP ☐ 0.5T SOLID STATE RF GRFD ☐ 0.35T ANALOGIC RFAMP ☐ 0.35T Teli RF Amp ☐ 3.0T ASTEC RF Amp ☐ 1.5T SRFD2 RF Amp
	✓ Governing Body	☐ iec ☐ mhw1 ☐ mhw2 ☐ fda2 ☐ Special 1 ☐ Special 2
	✓ CERD (KHz)	☐ MGD 250
	✓ ISO Vector Z (also record actual site value)	☐ 12520 Non-Magnishield Magnet ☐ 15250 Magnishield Magnet ☐ 8981 Cx/LCC Magnet ☐ 8689 nMR Magnet ☐ 8500 MFO Magnet ☐ 8689 SP Magnet
	✓ Magnet Enclosure	☐ WideOpen (tight fit) ☐ Pre-WideOpen (box style)
	✓ PAC Type	☐ PAC 1 ☐ PAC 2
Edit Patient	Edit Patient Data Time Limit	

TAB	ENTRY (✓ - Vital Info)	VALUE
Data	Max Images Installed Per Update	
	Log File Header	Log File Header (Default)
	Date Format	__ Month/Day/Year __ Day/Month/Year
	Data Base Century	__ 1900
Set Exam Number	✓ Exam Number (1-49999)	
	Diag Exam Number (50000-65535)	
Set Disk Limit	Set UDDM (60-100%)	
DASM	✓ Install DASM?	[] No [] Yes
	✓ DASM Interpolation	__ Linear __ Cubic
	✓ Laser Camera Type	__ 3M952 __ 3M XL or HQ __ 3M DryView __ AFGA MIN __ KODAK __ DUPONT FUJI KONICA
VCR Setup	✓ VCR Format	__ NTSC __ PAL
HIS/RIS – MR Host	Hospital Name	(Same as in “System Configure” tab)
	Suite ID	(Same as in “System Configure” tab)
	Host ID	(Same as in “System Configure” tab)
	✓ Local AE Title [Hostname] (1-16 Chars)	
	Local IP Address [Host IP]	(Same as in “Network” tab)
	✓ Local Port Number [Host Port] (0-65535)	
	Netmask	(Same as in “Network” tab)
	✓ DICOM Network Timeout	__ Slow (300secs) __ Medium (200secs) __ Fast (30secs)
	✓ Default Router IP Address	
	✓ Station Name (1-8 Chars)	
HIS/RIS – Modality Worklist	Current Patient Location	[] No [] Yes
	Special Needs	[] No [] Yes
	Pregnancy Status	[] No [] Yes
	Referring Physician	[] No [] Yes
	Request Contrast Agent	[] No [] Yes
	Contrast Allergy	[] No [] Yes
	Medical Alert	[] No [] Yes
	Additional Patient History	[] No [] Yes
	✓ MLW AE Title (1-16 Chars)	
	✓ MWL IP Address	
	✓ MWL Port Number (0-65535)	
	✓ Patient Name Delimiter	[] No [] Yes
HIS/RIS – Modality PPS	✓ PPS AE Title (1-16 Chars)	
	✓ PPS IP Address	
	✓ PPS Port Number (0-65535)	
Set Time/Date	Time Zone	

TAB	ENTRY (✓ - Vital Info)	VALUE
--	Modem Type or Broad Band Connection (Not in Install tab; look at modem)	

Check (✓) the Options installed at this Site on the System Options Table 1-2 below. After loading software and doing Restore Info, verify that all the options are still available to the customers. If any keys are missing, reactivate them using the option key floppy, or MOD.

TABLE 1-2
SYSTEM OPTIONS DATA TABLE

	SITE OPTIONS		SITE OPTIONS
	Echo Planar Imaging		Functool
	Fast Gradient Echo		Voxtool
	Cine		Interavascular Imaging
	Fast Spin Echo & FLAIR		Clariview
	Time of Flight		Multislice Multiangle
	Phase Contrast Vascular Imaging		Spectro Analysis GE
	Research		Performed Procedure Step
	Proactive Service		High Speed
	Research PSD		fgret
	Spectroscopy/PROBE		fgret Real Time
	Videoconferencing		Spiral Hi-Res
	SGD_EchoSpeed		Spiral Real Time
	DW EPI		iDrive Pro Plus
	FLAIR EPI		Probe 3D Brain
	SPECIAL		Ultrashort TR
	Smart Prep		T2 Breathhold
	SSFSE		SSFSE MRCP
	Three Plane Localizer		T1 Breathhold
	HIS RIS (Modality Worklist)		Line Scan Diffusion Imaging
	E3D TOF		Image Filter
	FSE_XL		ACGD Plus
	Bloodsupp		Probe 3D Prostate
	Fastcine		Fluoro-Triggered MRA
	Tagging		MRCP3
	Sgdperf		PSIR
	DW-EPIEZ		Dynamic R1
	iDrive Pro		Fiesta 2D
	iDrive		Fiesta 3D
	SmartPrep2000 Upgrade		Delayed Enhancement
	Probe2000 Upgrade		ASSET
	Multi-Nuclear Spectroscopy		Probe RRESS

1-4 SaveInfo and Saving System Information Entered In The Install GUI

1-4-1 SaveInfo

A SaveInfo must be done.

1. Insert the SaveInfo MOD or a new MOD into the drive.
2. Click on **[Save Info]** from the Install menu. The system opens the window called Xexec. (If the MOD has not been used, you will see a message regarding the use of the MOD for "Raw Data Storage"; choose **[Overwrite]**).
3. When asked: Do you really want to save configuration files to the Info MOD? [y|n], type: **y <Enter>**.
4. When SaveInfo is complete, you can try the following command to give you an indication that the SaveInfo was successful. Open a C Shell on the Service Desktop. Login as *root* by typing: **su root**, then enter password **operator**.

Type the following:

tar vtf /MOD_DRIVE <Enter> (there is a space between vtf and / character)

*Hundreds of files that were backed up on the MOD should scroll by at this point. This will be a good indicator that the SaveINFO was successful. Note: to view the files a page at a time you can type: **tar vtf /MOD_DRIVE|more <Enter>** (the vertical character "pipe" followed by the word "more" will display a page of output at a time; use the space bar to continue paging).*

Also as an indication that site protocols were saved successfully, you should see the something similar to the following (protocol names will not match, this just an example)

```
rw-rw-rw-  0/0          0 Jul 28 14:26 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/chest/.MrProtocol.chest.cfg
rwxrwxr-x  0/0  dir          Aug  7 10:17 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/
rw-rw-rw-  0/0          9472 Jun 28 13:20 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/.MrProtocol.head.cfg
rwxr-xr-x  64/32116      298496 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/1.0T_destroyer_head_A
rwxr-xr-x  64/32116      587898 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/Sr20PerformSpecs
rwxr-xr-x  64/32116      298602 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/destroyer_head_A
rwxr-xr-x  64/32116      132142 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/head_2d_thick
rwxr-xr-x  64/32116      22059 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/head_2dfast_cmon
rwxr-xr-x  64/32116      16571 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/head_2dfast_fcOFF
rwxr-xr-x  64/32116      16572 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/head_2dpc_fcOFF
rwxr-xr-x  64/32116      16592 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/head_2dtof_fcOFF
rwxr-xr-x  64/32116      27806 Jul 28 14:45 2000
usr/g/MRINFO/holdingtank/usr/g/protocols/site/head/head_3d_thick
.....
```

If site protocols are missing, please call OLC for support and do not proceed.

5. At top left corner of the Install GUI, select **<File>** then **<Save GI Configuration to MOD>**.

Note

The "Save GI Configuration to MOD" creates a copy of the information already entered in the GUI tabs for later use to update the tabs during a software reload. The <Save GI Configuration to MOD> and SaveInfo can be saved on the same side of the Info MOD. A separate partition is built for each.

6. Remove the MOD and write release version, the system name, and the date on it.

1-4-2 SaveInfo Backup File

Beginning with software release Signa 9.1 and 10.0, whenever a SaveInfo is made a backup file is created of all the files saved to the SaveInfo MOD. This backup file is located in /usr/g/service/log/saveINFO.tar.gz and can be used to manually update configuration files in the advent the SaveInfo MOD becomes corrupted. It is suggested that a backup copy be kept by using FTP to move the file to the service laptop or other machine and saved. See Appendix F for more information on this file.

1-4-3 SaveInfo Log File

Beginning with software release Signa 9.1 and 10.0, SaveINFO now creates a log file in /usr/g/service/log directory called saveINFO.log. This file will list all the files that saveINFO was expecting to save but could not find on the system and hence could not save. Some of the files may be specific to TWIN or HFO or MGD systems and in those cases the log file will have a comment to that effect so that the FE is not confused. It is not required but is a good practice that this file is checked after a saveInfo to insure all files have been successfully saved.

During the SaveInfo, a backup copy of the tar file saved to MOD is also saved to the systems hard drive in directory /usr/g/tmp. The backup file is named, saveINFO.tar.gz. In the event that a SaveInfo MOD would become corrupted, the saveINFO.tar.gz can be unzipped using the gunzip command. (On the command line type: gunzip -h for help using gunzip). Copy the resulting saveINFO.tar file to a new MOD and do restoreInfo as required.

1-4-4 Record Site Coils

All GE coils will be included in the Site Coils List. You will want to remove the GE coils that your customer does not have after the load, so record the site coils currently listed in the Coil Config File as follows.

1. From the Service Desktop, select **[Utilities]**, then **Config File Manager**. Select **Coil Config File**.
2. Record all coils below that are listed under **Site Coils**.

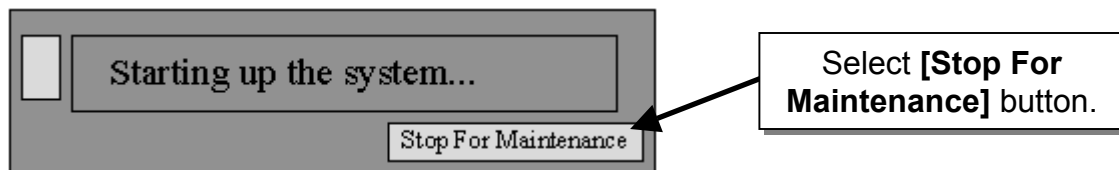
2- HOST OPERATING SYSTEM SOFTWARE LOAD

Initial Conditions:

- All images archived (Section 1-2).
- System information recorded (Section 1-3); SaveInfo performed (Section 1-4).
- System disconnected from Hospital Network.

Procedure:

1. Signa will only support the use of **ONE** 18 Gigabyte disk drive in the lower drive bay. No other combination of drives with the 18 Gigabyte drive is supported.
2. If the system is not up and running, switch on power at this time and continue with step 2. If the system is already running, go to the Service Desktop and select **[Shutdown]**. When prompted, click on the **[Restart Button]**.
3. Stop the reboot process; click on the **[Stop For Maintenance]** button when it appears.



4. Click on the **[Enter Command Monitor]** button when it becomes available.
5. Insert the **IRIX 6.5** CD labeled “**Boot/Update**” into the CDROM Drive on the workstation (right side). On the command line type the following (*user responses are bolded text*):

Boot the CD by typing the following at the command prompt:

```
>> boot -f dksc(1,3,8)sash64 dksc(1,3,7)stand/fx.64 --x <Enter>
```

```
fx: "device-name" = (dksc) <Enter>
.
fx: ctrlr# = (0) <Enter>
.
fx: drive# = (1) <Enter>
.
fx> r/ro <Enter>
.
fx/Repartition/RootDrive: Type of Data Partition = (xfs) <Enter>
.
fx> continue? yes <Enter>
.
fx> l/sy <Enter> (This is a lower case l , /, lower case s, and lower case y.)
```

.
fx> **exit** <Enter>

6. The system will reboot. Select **[Install System Software]** button. Choose, **[Local CD-ROM]** (the default), select **[Install]**. Then select the **[Continue]** button.

Note

If you are working with new or never used (Unformatted) disk drives, you will be told by the system during this reboot that the system was unable to mount the "/ root" file system. Without a format, the file system (Directory "/ root" being one of file structures) cannot be built. The system will stop with the statement: "press Enter to invoke C shell csh:". Type the following:

1. ...press Enter to invoke C shell csh: **<Enter>**
2. # **mkfs /dev/dsk/dks0d1s0** (This is /dev/dsk/dks zero d one s zero.)
(The file system will be built at this time.)
3. # **exit** <Enter>
4. **Reset the computer.** The computer will begin to reboot.
5. **Restart this procedure at Step 2 on Page 9.**

7. Make new file system on /dev/dsk/dks0d1s0 [yes/no/sh/help]: **yes**<Enter>
About to remake (mkfs) file system on: /dev/dsk/dks0d1s0.
This will destroy all data on disk partition: /dev/dsk/dks0d1s0.
8. Are you sure? [y/n] (n): **y**<Enter>
9. Block size of filesystem 512 or 4096 bytes? **4096**<Enter>
Doing: mkfs -b size=4096 /dev/dsk/dks0d1s0
meta-data=/dev/rdisk/dks0d1s0 isize=256 agcount=8, agsize=128612 blks
data = bsize=4096 blocks=1028895, imaxpct=25
log =internal log bsize=4096 blocks=1000
realtime =none bsize=65536 blocks=0, rtextents=0
Trying again to mount /dev/dsk/dks0d1s0 on /root.

10. Continue as follows:
inst> **sh** <Enter>
.
mount /CDROM <Enter>
.
/CDROM/mr_prep <Enter>
.

Note

During the software load, there will be much screen output and many pauses of 10 to 30 seconds, no entry is required at these pauses. Approximate wait times are listed for any lengthy pauses, wait for the prompts as detailed in the following steps.

11. The system will run for about 4 minutes and then will prompt with:

```
"Please insert the First OS Disk (Pa) and then press <enter>"
```

Put the OS CDROM Disk 1 of 2 into the CDROM Drive. Wait for 20 seconds, or until the amber light on the CD Drive stops blinking, then push the **<Enter>** key.

12. After an additional 3-6 minutes, the system will again prompt with:

```
"Please Insert the Second OS Disk (Pb), then press <enter>"
```

Put the OS CDROM Disk 2 of 2 into the CDROM Drive. Wait for 20 seconds, or until the amber light on the CD Drive stops blinking, then push the **<Enter>** key.

Loading the rest of the Operating System off the second disk and the system completing the software load will take about 30 minutes and will not require any user intervention.

13. When finished, the system will stop at the "#" prompt. Type the following to continue.

```
.
# exit <Enter> (to exit back to the original inst> prompt.)
.
.
Inst> exit <Enter>      there is a delay of about 30 seconds, then:
.
Automatically reconfiguring the operating system..
.
.
.....Ready to restart... y <Enter> (Restarts the system)
.
```

The system will finish some processing and reboot. The OS load portion of the Software Install is complete.

Important: If at any time during the OS load a problem occurs that causes you to stop the install, correct the problem then reset or power cycle the computer and begin the OS install again from the start. This is necessary to insure a complete load. If this does not fix your problem it may be necessary to do a low level format on the system drives. Refer to Appendix C for this process.

14. Continue to Section 3 of this document for the MR_Apps software load.

3- MR_APPS SOFTWARE LOAD

1. When the system is finished rebooting, log in as **root**. There is no password at this time. You will first see a startup box that will asks for the MrpApps CDROM. See Illustration 3-1.



GUIDED INSTALL LOAD SCREEN
ILLUSTRATION 3-1

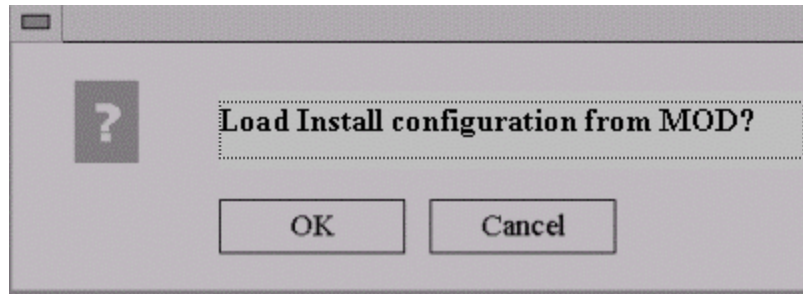
2. Remove any CD from the CDROM drive and insert the MrpApps CDROM into the CDROM drive. Wait 20 seconds or until the amber light on the CDROM drive stops blinking and click on the **<Load>** button. The system will then respond with a popup that says "Make sure MrpApps is in the CDROM drive". Insure the CD is in the drive and again click the **<Load>** button. The Guided Install will now start.

Note

Help documentation is built into the Guided Install GUI. Push the **<Help>** button on the tab you need information for in the GUI. When the GUI first starts, press the **<What's New>** button for information on using the GUI.

3. On initial startup, a popup box will appear over the Guided Install interface describing the tool and its use. After reading the information push the **<OK>** button to continue.
4. Next, a popup box (see Illustration 3-2) will appear over the Guided Install interface.

Important: The last Install GUI configuration can only be installed if it was previously saved using the **<Save GI Configuration to MOD>** option per Section 1-4-2, Save GI configurations.



INSTALL CONFIGURATION POPUP SCREEN

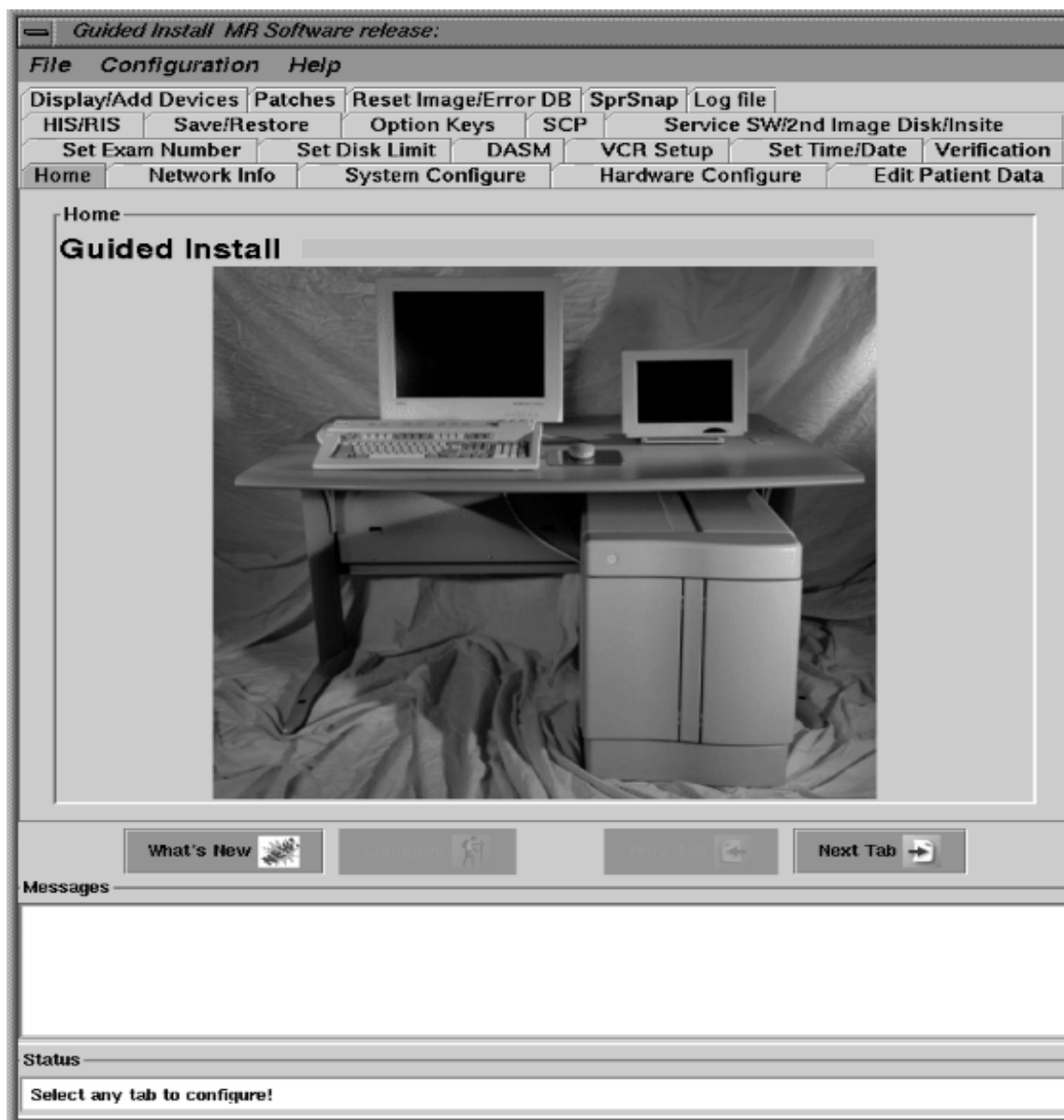
ILLUSTRATION 3-2

5. Insert the SaveInfo MOD (created in Section 1-4-3, Save GI Configurations) then push the **<OK>** button to reinstall your last GUI install configuration values. *(The normal SaveInfo files are not restored at this point in the software load. They will be restored in step 9.)* You will verify the correct values were restored by checking each tab in the next step.

Note

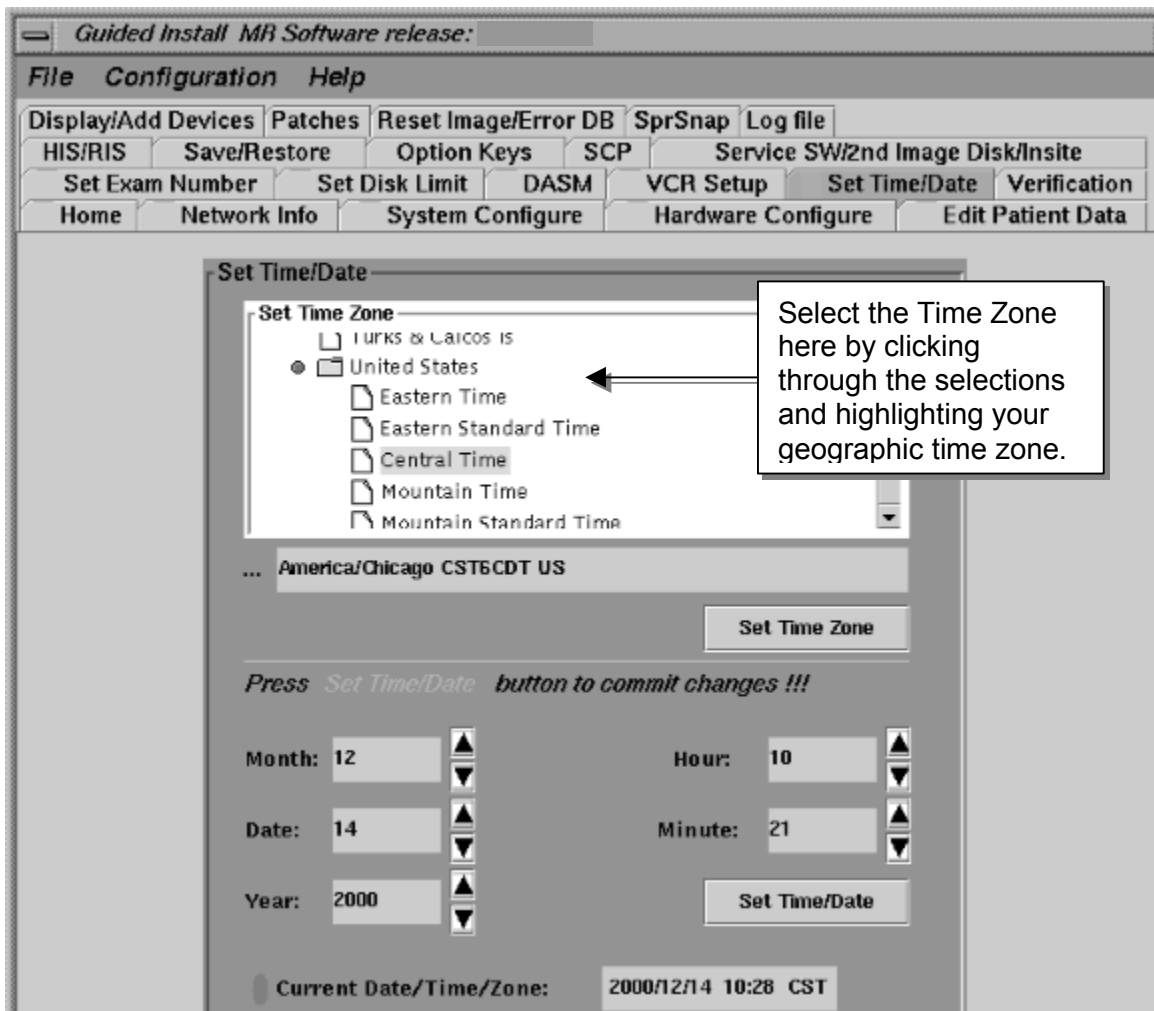
If you did not perform Section 1-4-2, Save GI Configurations push the **<Cancel>** button to continue manually with the install.

6. The Guided Install home screen is displayed. See Illustration 3-3. It is now OK to go through the tabs using the mouse.



GUIDED INSTALL - HOME SCREEN
ILLUSTRATION 3-3

7. Press the **<Next Tab>** button to move through active tabs from "**Network Info**" to "**Set Time/Data**". Fill out all empty entries or make any updates as appropriate. This information should have been saved in Table 1-1 of this document. When you reach the "Set Time/Date" tab, go to the "Set Time Zone" box and use the selections to set the time zone your system is in. See Illustration 3-4.



GUIDED INSTALL - SET TIME/DATE SCREEN
ILLUSTRATION 3-4

8. In the **Hardware Configure** tab, there is an entry for CERD that has a confusing number of entries. For Signa 10.x MGD systems, the correct entry is "**MGD 250**".
9. Proceed to the **Verification** tab. Once in the Verification tab, look at the pass or fail status of the entries. If any entry has a failed status, the **<Install Software>** button will not be active. Failed entries will be due to missing entries, or entries not in the correct format. Go to the tab with the error and correct it, then push the **<Install>** button to open the Verification tab again. If all entries pass verification at this point, push the **<Install Software>** button again to begin the MR Apps load. The MR Apps load will take 20-30 minutes, depending on the machine being loaded.

Note

It is not necessary that any one tab be filled out in any particular order. Just all required configuration tabs must be filled in or the install will not continue. The Verification Tab will list any formatting errors as well as current settings. See Illustration 3-5 for an example of a failure in the Verification Tab in networking (in the example, the Host IP Address is an incorrect format).

Guided Install MR Software release:

File Configuration Help

Display/Add Devices Patches Reset Image/Error DB SprSnap Log file

HIS/RIS Save/Restore Option Keys SCP Service SW/2nd Image Disk/Insite

Set Exam Number Set Disk Limit DASM VCR Setup Set Time/Date Verification

Home ? Network Info System Configure Hardware Configure Edit Patient Data

Verification Failed! Configure tab(s) Go to Tab

Status	Parameters	Value
	Time Zone	America/ChicagoCST6CDTUS
	Field Strength	1.5T
	Package Name	ECHOSPEED
	Gradient Type	8651
	Resonance Module	CBM
	Magnet Serial Number	Dxxxxx
	Table Limit	31750
	Line Frequency	60Hz
	RF Amp Type	Tube Type
	Governing Body	fda2
	CERD (kHz)	CERD 1/4 C
	ISO Vector Z	8981 CX / L
	Isocenter Vector along Z-axis	8981
Fail	Host IP Address	3.7.25.
	Netmask	0xffffc00
	Hostname	ts
	Domain Name	mirceg
	YP(NIS) Service	On
	IWS PC IP Address	3.7.27.9
	Magnet Monitor IP Address	1.1.1.0
	FE Laptop IP Address	192.168.4.2
	TPS Subnet	216.33.27
	Default Router IP Address	
	Power Monitor PC IP Address	
	Hospital Name	ts

Messages

7:37:28; GUI Warning ! Invalid entry. This field cannot begin or end with a dot----- Host IP Address : 3.7.25.

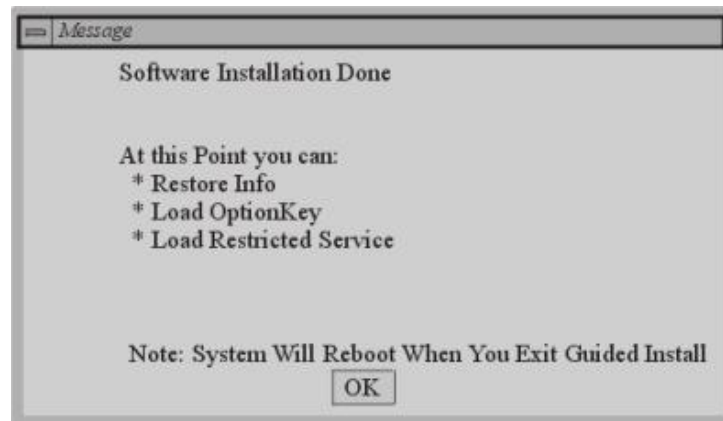
Status

Enter IP Address (eg. 2.33.44.55)

Callout Box: Errors are shown in red on color monitors, or light gray in Monochrome monitors. Error details are shown in the message area at the bottom of the Install GUI. This failure is due to an incorrect IP address format. The correct format is: xxx.xxx.xxx.xxx. The <Help> button on the Network Tab gives the correct format.

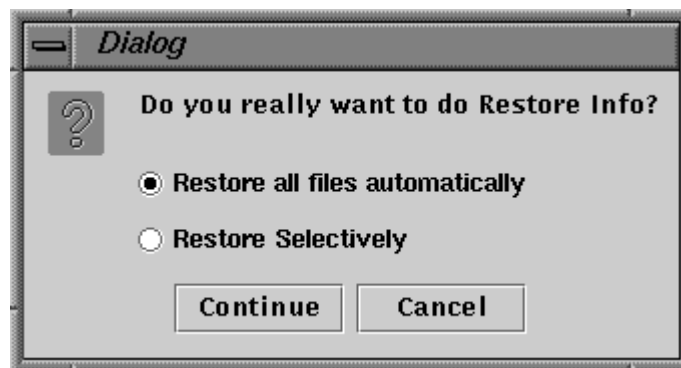
GUIDED INSTALL - VERIFICATION SCREEN
ILLUSTRATION 3-5

10. Once the MR Apps load is done the popup in Illustration 3-6 is displayed. Click on the **<OK>** button to close the popup.



GUIDED INSTALL – SW DONE SCREEN
ILLUSTRATION 3-6

11. At this point the GUI will change from the Install phase to the Configure phase. If a SaveInfo info MOD is available for this site, proceed to the **Save/Restore** tab and select **[Restore Information]**. If no SaveInfo MOD is available, go to each tab in the Install GUI and fill in or update all system information manually. A help button is available on each tab, which will open a popup box with information on the required data and its format for that page.
12. A popup will ask if you want to restore all files or selectively restore, see Illustration 3-7. Select **Restore all files automatically**, then **[Continue]**.

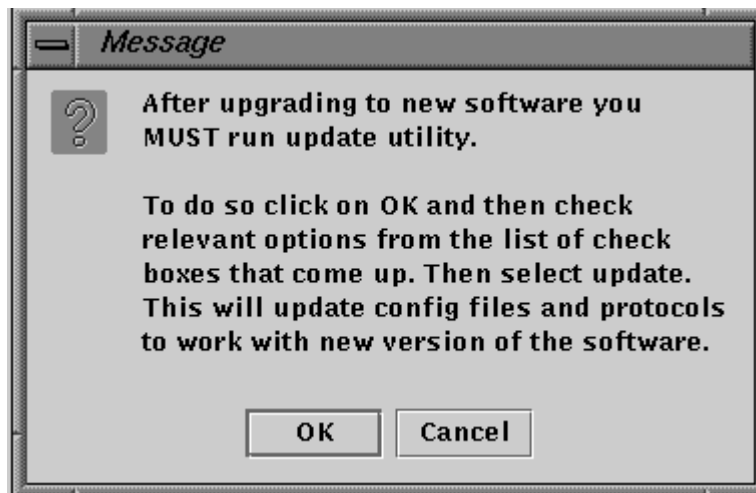


RESTORE OPTION POPUP
ILLUSTRATION 3-7

Note

If RestoreInfo fails, the CDROM reader will automatically eject the media. Exit install and allow system to reboot. When the system is back up, re-enter *Guided Install* and attempt to RestoreInfo again.

13. A “RestoreINFO in Progress” window will scroll filenames as they are being restored.
14. When the RestoreInfo is finished a popup will open telling the installer that to run the update utility. See Illustration 3-8.



GUIDED INSTALL – UPDATE UTILITY MESSAGE
ILLUSTRATION 3-8

15. Push the **<OK>** to continue. The Restore Manager menu opens (see Illustration 3-9).



GUIDED INSTALL – RESTORE UTILITY
ILLUSTRATION 3-9

Make sure to check the following:

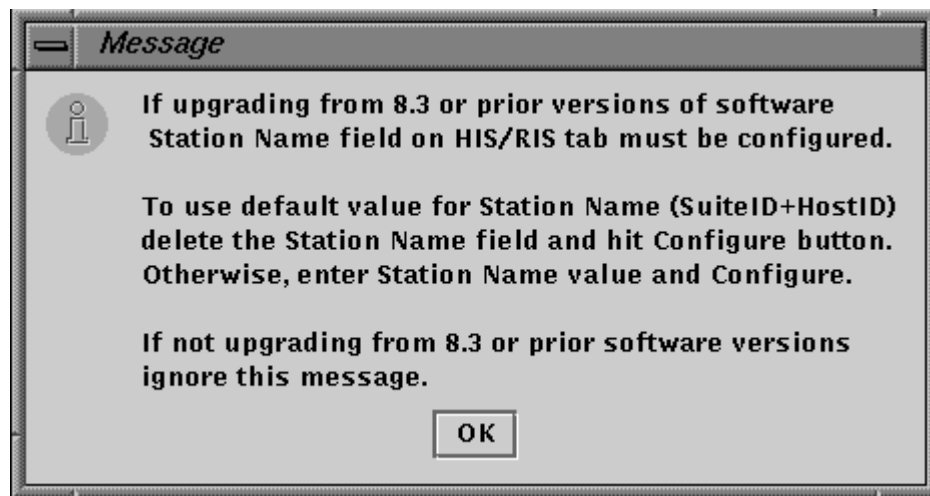
- a. Protocols
- b. CoilConfig.cfg
- c. MRconfig.cfg
- d. GradientConfig.cfg

16. Click on **<Update>** button when update is finished Click**<Quit>**.

A logfile of changes are made by the Restore Manager and is stored at:
/w/config/restore.log

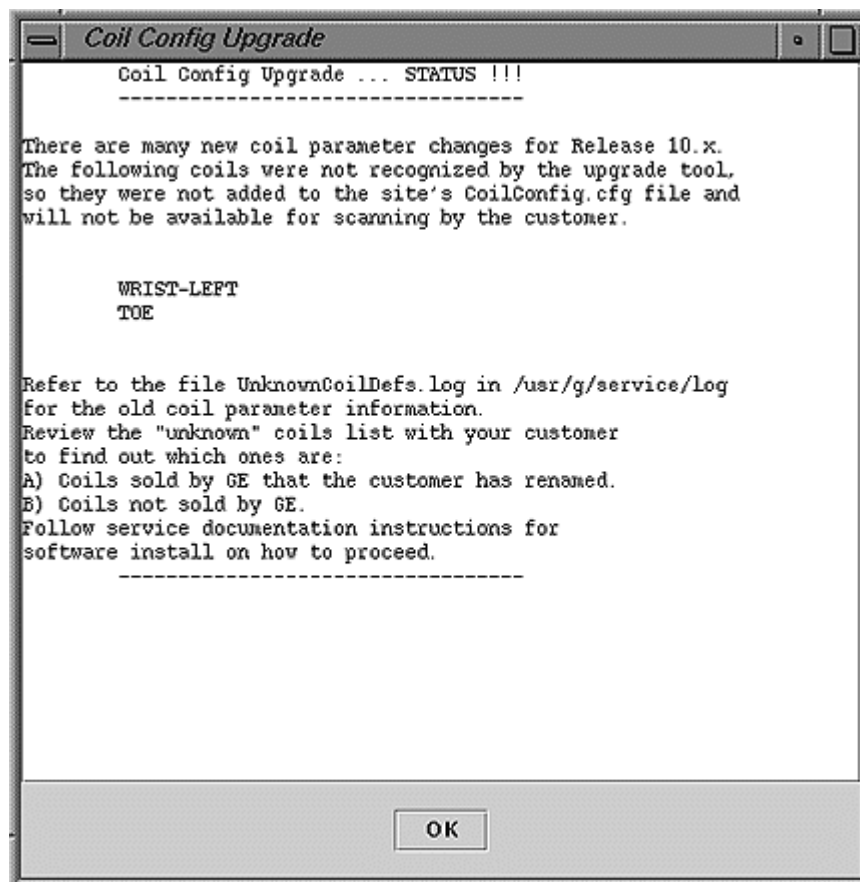
17. Select **[OK]**.

18. The following Message will appear. (See Illustration 3-10) Select **[OK]**.



8.3 UPGRADE MESSAGE
ILLUSTRATION 3-10

19. Beginning with Signa 10.x coil parameters have changed. For upgrades from 8.x or 9.x, a coil updated will run to add the new parameters to the exciting coil definitions. The success of the update is dependent on recognition of coil names. If no discrepancies are found continue to step 20. If any coils are not recognized during the update, the error message shown in Illustration 3-11 will popup.



COIL CONFIG UPGRADE STATUS MESSAGE WITH UNKNOWN COILS FOUND
ILLUSTRATION 3-11

If unknown coils are found, finish the software load then go to section 9 of this document for help in adding these unknown coils.

20. Use the cursor to select **<File>** and then **<Quit>** at the top left of the Install GUI. When the GUI shuts down, the system will reboot and all the changes made during the restore process will be set.
21. After the system has rebooted, log in as **root**. Password is **operator**
22. Select **[Yes]** to start the Install GUI.

Note

(Restricted Service Software is only available for GE use; Advanced Service Software is only available to sites with a valid Advanced Service Package Limited License.) GEMS Field Engineers loading GEMS Restricted Service Software should also configure InSite Interactive at this time (see Illustration 3-12). Follow the instructions given in popup windows as presented. For a more complete discussion on configuring the InSite package, see Appendix E of this document.

23. Insert the Service Software CD and click the Service S/W Tab, then select Restricted or Advanced software load button. The system will respond with a popup as shown in Illustration 3-12. Read the proprietary statement and continue by clicking the "1,2,3" buttons in order.

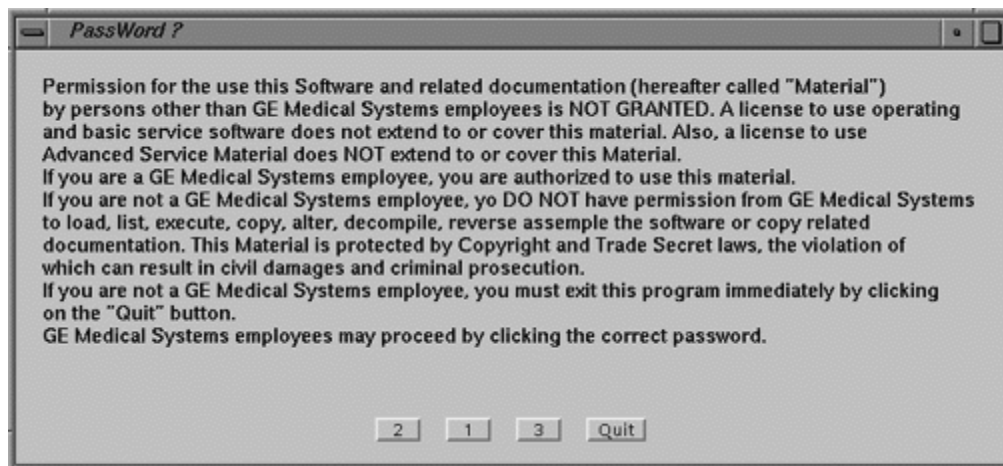
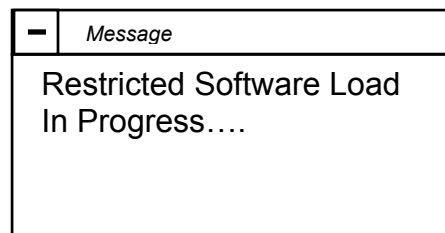


ILLUSTRATION 3-12
PROPRIETARY PASS WORD SCREEN



24. This load will take about 3 minutes to complete.

InSite Interactive Platform Configuration

Information ProDiags Health Page **Device Connection** InSite Checkout

Connection Device Configuration
Last Update: Fri Aug 13 15:50:42 CDT 1999

Device Connection Type **Modem**

Select the InSite Connection Type above, or select the Modem settings given below and Press

APPLY - To update the modem and modem configuration files with model and country defaults.

CUSTOM - For other modem settings.

Dial-out Prefix: 8, **Tone**

Dialing mode: **Tone**

Modem Type: **USRobotics Courier V.34 or V.Everything** Country: **Default - All Others**

CPU Serial Port Name: **/dev/ttyf1** CPU Serial Port Speed: **115200**

APPLY **CUSTOM >>**

Status: IIP Configuration Tool Ready for Input... **EXIT**

ILLUSTRATION 3-13
INSITE INTERACTIVE PLATFORM CONFIGURATION SCREEN

Note

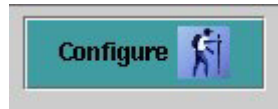
When configuring the System Health page of **Configure InSite Menu**, you are able to enter multiple email address where reports are sent. The instructions on that tab are incomplete. You must push the **<Enter>** key after every email address entered.

Note

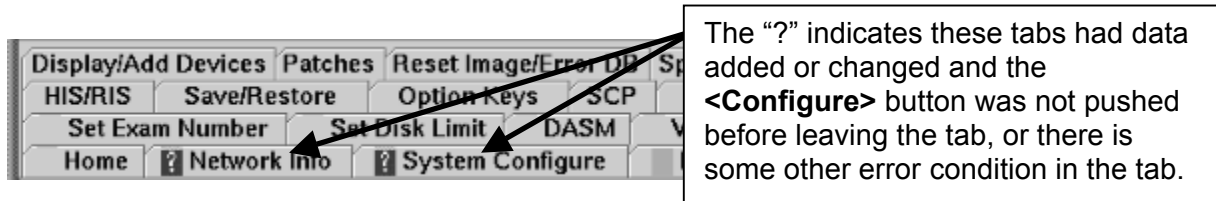
There are two software MOD keys that must always be reloaded. These two MOD's (two sides on one MOD and one side on the other MOD to load) contain the LX Tools software that is shipped with every system.

25. Turn YP on only if the site's external network requires it. If YP is turned on, the system will take longer to boot (2 to 4 minutes). Be aware there is a small delay that may not be seen on the sites.

26. After information as been restored from MOD or changed in a tab, and configuration updates have been run, push the **<Configure>** button on the bottom of the tab to apply the configuration.



If the **<Configure>** button is not pushed to commit changes to a tab, or an error is present in a tab, a question mark will be seen on the tab, as shown in Illustration 3-14.



CONFIGURATION PROBLEM EXAMPLE

ILLUSTRATION 3-14

27. Perform a SaveInfo to save the latest changes by inserting the site SaveInfo or blank MOD into the drive then going to the Save/Restore Tab and pressing the [Save Information] button.
28. Insert the your normal SaveInfo MOD or a new MOD into the drive if not already in the drive.
29. At top left corner of the Install GUI, select **<File>** then **<Save GI Configuration to MOD>**. This will save information that has been entered in all the edit fields in the GUI.

Note

The “save to MOD” process does not create a complete SaveInfo disk. It only creates a copy of the information already entered in the GUI tabs for later use to update the GUI tabs during a new install. (*SaveInfo uses the Save/Restore tab.*) The **<Save GI Configuration to MOD>** and SaveInfo can be saved on the same MOD. A separate partition is built for each.

30. Exit the install GUI by selecting **<File>** then **<Quit>** from the top of the interface. If any error conditions still exist, you will be warned that no changes will be made. If no error conditions are found, the GUI will save user parameters.

31. Reconnect the following hardware at this time if it was disconnected at the beginning of the install:
 - Network Interface
 - DASM
32. If the system was using two 9 Gigabyte disk drives the second drive should be reinstalled at this time.
 - A. Log out of root and shutdown the computer and remove power.
 - B. Reinstall the second disk drive into the middle drive slot.
 - C. Power up the system and log into root and start the configuration GUI. See Illustration 3-14.
 - D. Go to the **Service SW/2nd Image Disk/Insite** tab and push the **[Install 2nd Image Disk]** button. The disk will be configured in within seconds. See Illustration 3-15

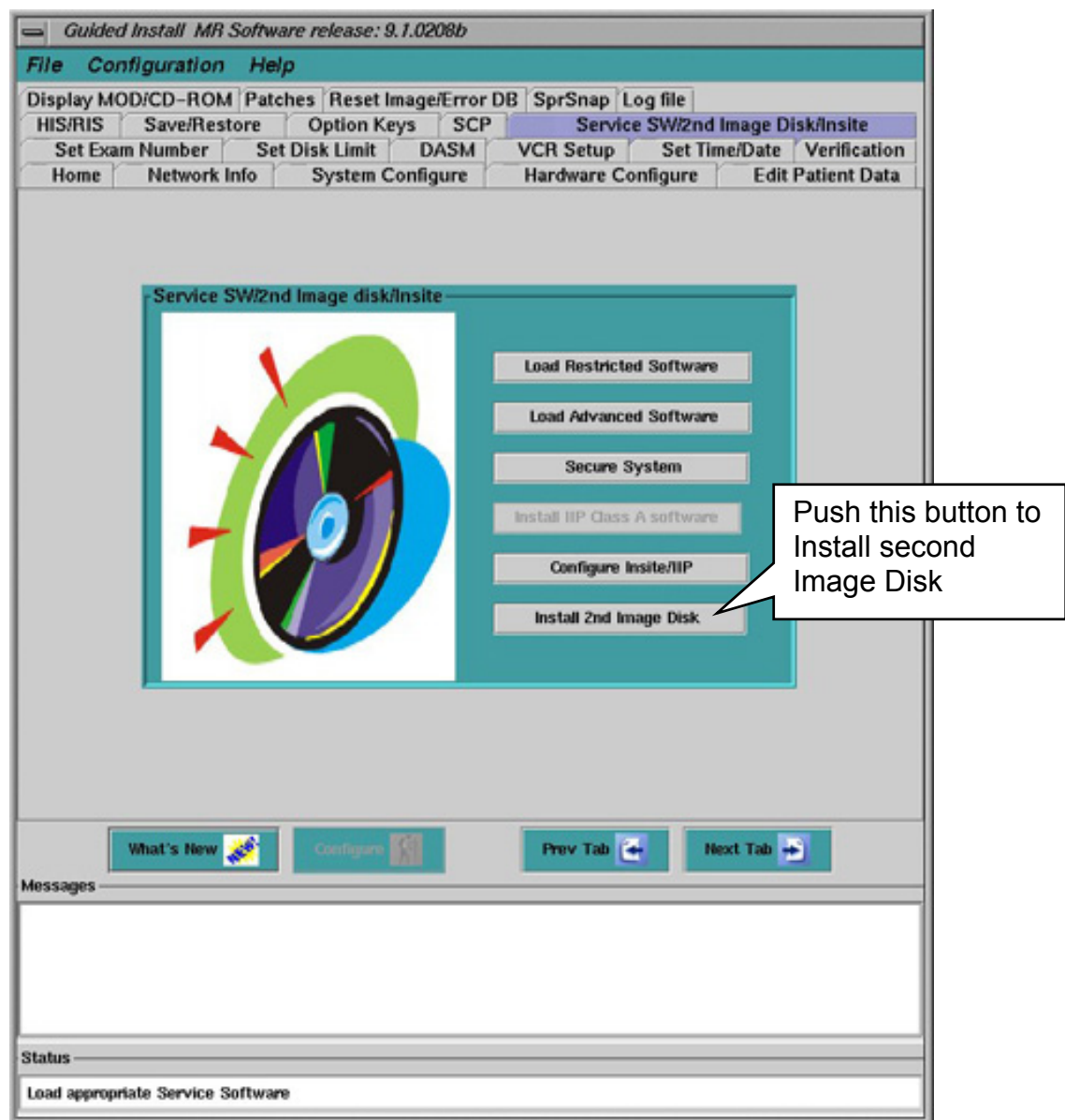


ILLUSTRATION 3-15
INSTALL 2ND IMAGE DISK BUTTON

33. Exit the Configuration GUI, log out of root and reboot the system.

4- VCR INTERFACE OPTION

If your system does **not** have the VCR I/F Option proceed to Section 5. Systems that have the VCR Interface Option will have to follow the steps in this section to load the video driver because it is not loaded with the operating software load. Follow these steps to install the video driver software:

1. From the log in screen, log in as **root**. Password is **operator**. You will be asked once entering the root desktop if you want to start the Install GUI. Select **[No]**.

1. Insert the IRIX 6.5 Boot/Update CDROM into the CDROM drive.

2. In the winterm window type: **mediad <Enter>** (to mount the CDROM drive).

3. Type: **cd /CDROM/opt <Enter>**

4. Type: **./vidsetup <Enter>**

This starts the install script and a lot of screen output will scroll by in the window, and it will be apparent that file installs/removals are taking place. Then the last output to be displayed is below:

```
Reboot to start using the reconfigured kernel.
*****
**                                                    **
**                                                    **
**      Remember to reboot the system before         **
**      attempting to use the video interface         **
**                                                    **
*****
{ sdc@name }
```

5. Type the following to verify the video driver was loaded:
showprods -D1 <Enter>

examine the products that are loaded and ensure the following is listed:

```
evo <date installed> OCTANE Personal Video Execution Environ,
6.5.4m
```

6. Press the spacebar until you get a prompt back.

7. Type: **mediad -k <Enter>** (to release the CDROM device).

8. Remove the Boot/Update CDROM.

Note

Do not reboot the system at this point to activate the video driver. The system will be rebooted after loading the patch software in the next section.

5- GAMMA SWAP & FRAME RATE, CONTRAST, & BRIGHTNESS & GRAYSCALE CALIBRATION

After a complete software load has been done, proceed to the specific Host Monitor Setup and Calibration procedure for your monitor type. Failure to perform the Monitor calibration will result in Customer complaints related to Image Quality and may cause unnecessary system troubleshooting.

To make sure that LCD monitor image quality is optimized, set up the following:

- Gamma Setting (Selecting the correct monitor type in Guided install will automatically install the correct gamma table for that monitor type.)
- Swap and Frame Rate
- Contrast and Brightness
- Grayscale
- DASM (If present)
- Camera (If present)



FOR 10.0 SOFTWARE: ALL GAMMA LOOK-UP-TABLES OR SINGLE GAMMA VALUES HAVE BEEN INCORPORATED INTO THE SIGNA SOFTWARE RELEASE.

DO NOT LOAD ANY GAMMA LOOK-UP-TABLES FROM THE SERVICE CDROM OR WEBSITE. DOING SO WILL CORRUPT THE SOFTWARE.

- For NEC 2010, 2010X LCD monitors use [[NEC 2010, 2010X Host LCD Color Monitor Calibration](#)].
- For Eizo L660 LCD monitors use [[EIZO L660-Host LCD Color Monitor Adjustment](#)].
- For NEC 1850X LCD monitors use [[NEC 1850X – Host LCD Color Monitor Calibration](#)].
- For NEC 1880SX LCD monitors use [[NEC 1880SX – Host LCD Color Monitor Calibration](#)].
- For MDW321 (B/W) & MDW321+ (Color) CRT monitors use [[MDW321\(B/W\) & MDW321+ \(Color\) CRT](#)].
- For monitors that don't match any of those previously listed then use [[LCD or CRT Monitor Cal & Gamma Setup](#)].

6- RECONFIGURING THE SYSTEM USING THE INSTALL GUI

If this is a new/upgraded Signa installation or if the system just needs to be re-configured between software loads, do the following to start the Configuration GUI:

1. With Signa up and running, go to the service desktop and push the **<Install>** button on the toolbar. A C-Shell will open and ask for root password ("**operator**" **<Enter>** is the default). The Install GUI will load and start.
2. Help documentation is built into the GUI. Push the **<Help>** button on the tab you need information for in the GUI. Push the **<Next>** button to move through tabs in order. Modify entries to match your system as required. When satisfied, push the **<Configure>** button at the bottom of the tab page or wait until all tabs have been modified and go to the "**Verification**" tab. Review changes made are correct, then push the **<Configure Tab(s)>** button to update the system (see Illustration 4-1).



VERIFICATION MENU – CONFIGURE TABS
ILLUSTRATION 4-1

3. Before exiting the GUI, insert the SaveInfo MOD or new MOD into the drive. Go to the top left corner of the Install GUI and select **<File>** and then **<Save GI Configuration to MOD>**. This process does not create a SaveInfo disk. It just creates a copy of the information already entered in the GUI tabs for use in the next software install.
4. Exit the install GUI by selecting **<File>** then **<Quit>**. If any error conditions still exist, you will be warned that no changes will be made before exiting.

Note

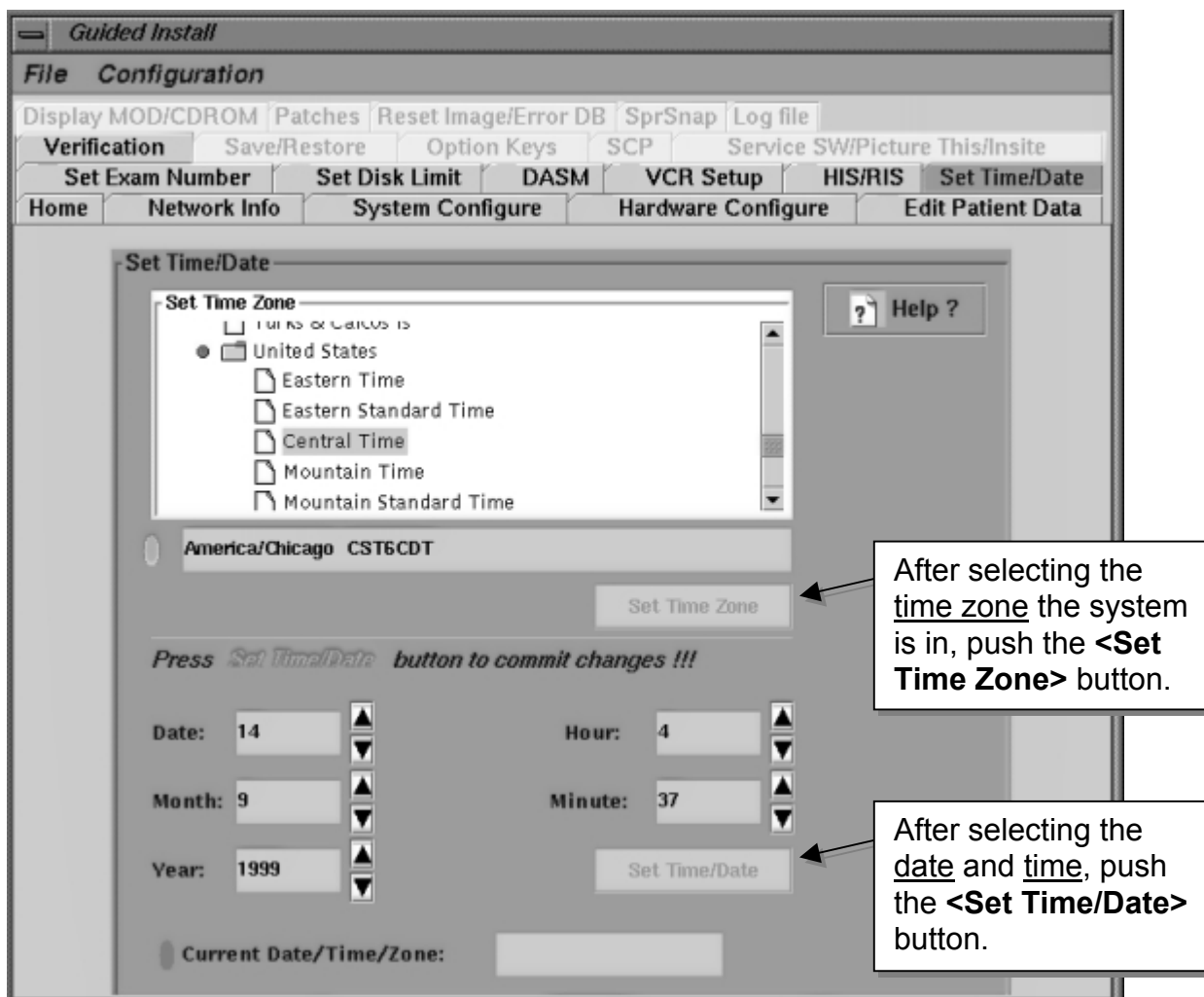
The GUI will not automatically reboot the system in configure mode. It will be up to the user to reboot after making configurations.

5. If changes were made to the system configuration, select **[System Shutdown]** and reboot the systems for changes to take effect.

7- SET TIME/DATE TAB

The **Time/Date** tab is the only tab that the Configure button will never be highlighted to set options. There are separate buttons for configuring this tab (see Illustration 7-1). If you need to re-configure the Host time and/or date at any time, do the following:

1. Select the Time Zone your system is located in, then push the **<Set Time Zone>** button.
2. Select the date and local time, then push the **<Set Time/Date>** button.



SET TIME/DATE MENU
ILLUSTRATION 7-1

8- SERIAL EXPANSION BOX FIRMWARE UPGRADE

This section should only be performed on systems upgrading from Signa 8.x or 9.x to 10.0 or higher. If the Serial Expansion Box is not at the proper revision, the MGD chassis will sometimes fail to boot causing TPS reset to fail.

8-1 Checking Firmware Revision

1. Log into the Octane computer as root or if already running in SDC, open a C-Shell and type **su -root <Enter>**
2. From a UNIX shell execute the following command to list information about the SCSI device in the system:

> scsistat <Enter>

3. In the output from the scsistat command, check for information about the SCSI terminal server. There are 4 currently known versions of the SCSI terminal server. Others may exist. Their scsistat information is:

a. Device 2 4 Comm CenData ST-1008S FW Rev: V6.4	← Must Update
b. Device 2 4 Comm CenData ST-1008S FW Rev: V7.2	} Other Firmware is OK.
c. Device 2 4 Comm DigIntl ST-1008S FW Rev: V7.2	
d. Device 2 4 Comm CenData ST-1008G FW Rev: V5.5	

If the SCSI Terminal Server information on your system matches (a), your expansion box firmware needs to be updated by following the direction in the next subsection.

8-2 Updating The Firmware

Do not perform this firmware upgrade unless you determine your SCSI terminal requires updating as shown in section 8-1.

1. Log into the Octane as Root if not already there or if already running in SDC, open a C-Shell and type **su -root <Enter>**.
2. Open a UNIX shell and change directory to /usr/g/bin.

> cd /usr/g/bin <Enter>

3. In the UNIX shell now enter the following command to update the SCSI Terminal Server firmware:

> /sbin/cdupdate <Enter>

4. The system will scan for STS devices and stop with a display of the available devices. There should only be one. Type **1 <Enter>** to chose that devise.
5. When prompted "Program ST1008S unit at buss 2 Target 4 (y/n)", type **y <Enter>**.

6. When prompted to “Enter Filename or <CR> to use default (st1008s.prm)”, Push the **<Enter>** key on the keyboard to use the default filename.

The firmware will be updated. Confirm the update was successful by performing the check in section 8-1 again.

9- RELEASE 10.X COIL CONFIGURATION UPDATES

This section should only be completed on:

- Systems upgrading from Signa 8.x or 9.x to 10.x or higher, and the system reported unknown coils during restore manager update of CoilConfig.cfg **or**
- Systems that you think are missing default parameters for new coils that vendors released to market after the last software release **or**
- Systems with default coil parameters that you think may not be correct.

Beginning with Signa 10.0, the number of parameters used in coil configuration has grown significantly. It is no longer a simple matter to configure new surface coils to Signa. As a result, a new process has been developed which auto configures known coils. The configuration is based off a known coil definition file that is keyed on known coil names.

Note

You should do a Save Protocols before beginning the coils configuration process. This will allow for recovery in the event that something goes wrong in the coil update process.

There are two main reasons why a coil may not be part of the coil definition file:

- A) Coils sold by GE that the customer has renamed from the GE standard name.
- B) Coils not sold by GE and or coils created by researchers.

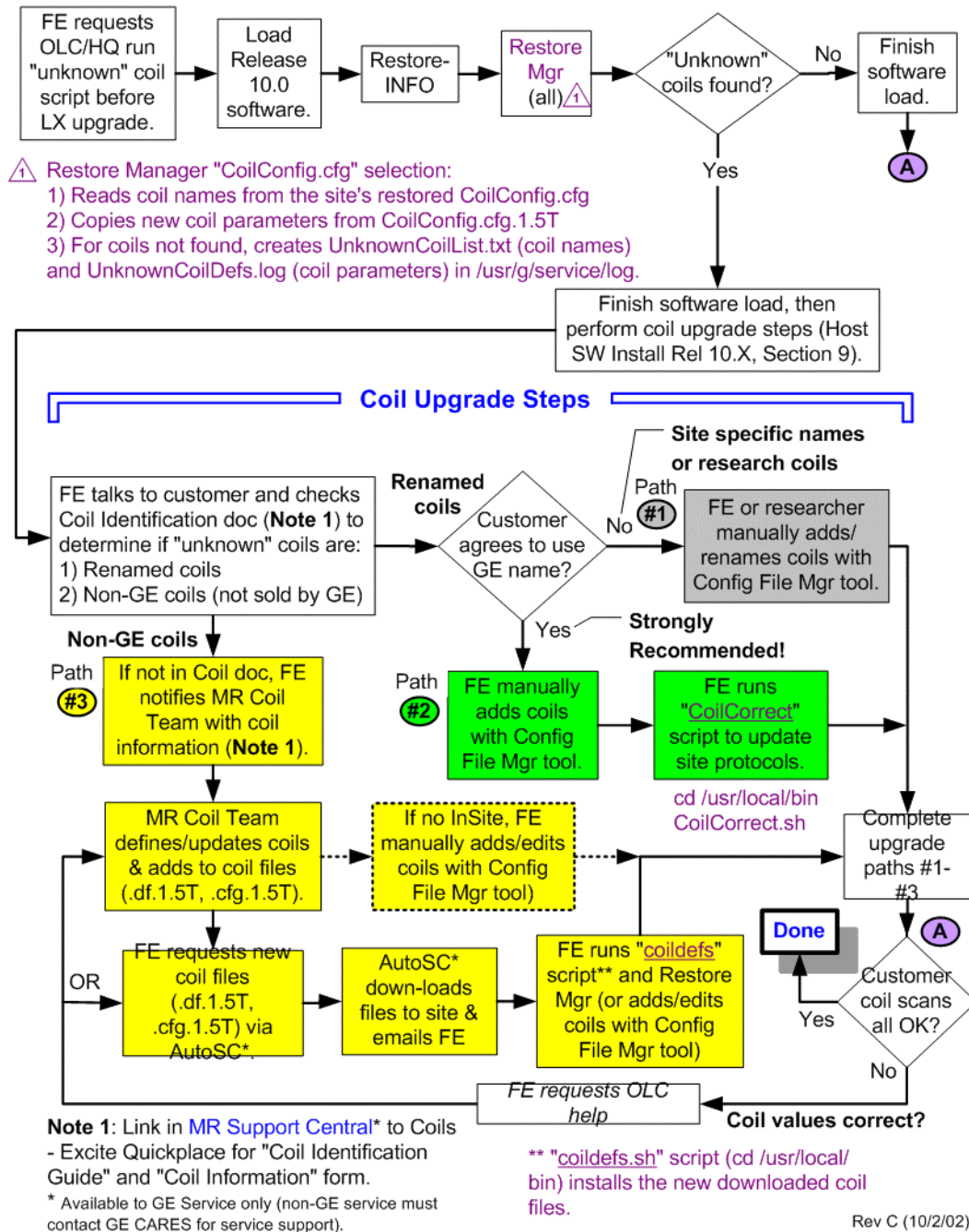
Once coils have been flagged by the system as being unknown, the unknown coil(s) will have to be added to the system by one of three methods before it can be used.

1. GE coils not using standard GE names that the customer does not agree to rename. These coils must be manually added/renamed using the Config File Manager tool. Non-GE or research coils are manually added using Config File Manager tool. All parameters will have to be updated manually.
2. GE coils that have been renamed are renamed back to their standard GE name. These coils must be manually added using the Config File Manager tool. A script is provided to update customer protocols if this step is used.
3. An update to the coil definition file is requested from GEMS HQ Coil Team. Once created the new file is downloaded and the coils are configured to the system using a script provided with 10.x software.

The first step to resolve unknown coil problem is to obtain the GE Coil Identification Guide. This document may be found on the MR Service Methods CDROM under the Coils page. The latest copy can always be obtained by GE Employees by going to [Support Central - MR](#) on the GEMS Intranet and downloading the document from under **Manuals → Signa Documentation → Coils – 10.X and above. Signa Infinity & Excite Technology → Q.P. 9.0X and 10.X Coil Listing → Coil Identification Guide.** This document can be used to help determine if unknown coils are renamed coils, new coils produced after the software was released, or non-GE (Not sold by GE) coils.

Once the unknown coils are identified, refer to Illustration 9-1 for the coil update process and determine what action should be taken.

RELEASE 10.X COIL CONFIG UPGRADE PROCESS (LX UPGRADES)



RELEASE 10.X COIL CONFIG UPGRADE PROCESS (LX UPGRADES) ILLUSTRATION 9-1

Sections 9-1 through 9-3 explain in detail the paths shown in Illustration 9-1.

9-1 Coils Are Manually Added To the Signa System

This section covers path (1) on Illustration 9-1. To view all the coils that have been flagged as unknown, go to the Service Desktop and Open a Cshell. Then type at the command prompt:

```
> more /usr/g/service/log/UnknownCoilList.txt <Enter>
```

Coils are manually added to the Signa system using Configuration File Manager. To start the Configuration File Manager go to the Service Desktop. If not already started, start the Service Browser, Choose the Configuration Tab and push the Configuration File Manager link on the left.

Add the additional coils as needed. Refer to Service document [Configuration File Manager \(Release MGD/10.x\)](#) section 3 for help in adding new coils.

9-2 GE Coils or Other Known Coils are Renamed Back to GE Standard Names

This section covers path (2) on Illustration 9-1. This section also assumes that known GE coils have been renamed to non-GE standard names and the customer has agreed to use the standard GE names again. The coils that couldn't be found on the original update will have to be added manually using the correct names. After the coils are added, existing customer protocols that use the non standard name can be updated via script to the new coil name. Continue as follows.

1. To view all the coils that have been flagged as unknown, go to the Service Desktop and Open a Cshell. Then type at the command prompt:

```
> more /usr/g/service/log/UnknownCoilList.txt <Enter>
```

2. Coils are manually added to the Signa system using Configuration File Manager. To start the Configuration File Manager go to the Service Desktop. If not already started, start the Service Browser, choose the Configuration Tab and push the Config File Manager link on the left.
3. Add the additional coils as needed. From the "Available Coils" list, select the added coil(s) from the "Site Coils" list (added to bottom of list), then click **[Edit Coil]**. Change coil name back to the GE coil name.

As an example, if the customer had a coil named "CHEST" that was using the GE TORSO coil, in Config File Manager add the CHEST coil back to the site list and then edit the coil and change the name to TORSO.

Note

Once finished it may take a few minutes for the Coil Config to update to complete (refer to status line at the bottom of Config File Manager window).

Refer to Service document [Configuration File Manager \(Release MGD/10.x\)](#) section 3 for help in adding new coils.

4. Once a coil name has been changed, update the customer existing protocols by doing the following.

Open a Cshell and type the following at the command prompt:

```
> cd /usr/local/bin <Enter>
> CoilCorrect.sh <Enter> (If only one coil needs updating) or use
> AllCoilCorrect.sh < Enter> (This will allow walk through multible
coils)
```

In the command window will appear a list of the coils that are currently used in the customer protocols. Enter the coil name you want to change. The next question will be what should the name be changed to. Give it the new name used in step 3 above, then a final verification is made and the scrip changes the coil names in the protocol.

Using the example given above, changing the customer coil name "CHEST" back to the GE name "TORSO". Illustration 9-2 shows the screens as seen on Signa. Illustration 9-2 is an example screen. The number of coils shown for your system may vary.

The coil CHEST is being used in protocols on this system.

```

@b11b}[21] cd /usr/local/bin
@b11b}[22] CoilCorrect.sh

Coils existing in Protocols

HRBRAIN      CS123      HEAD      Lumbar CTL      Thoracic CTL
IVARRAY      CTLBOT      HI RES BRAIN  TL34      USCTLBOT
BODY         CTLMID      LS45      TL345      USCTLMID
CHEST        CTLTOP      LS456      TORSOPA    USLS456
CS12         Cervical CTL LS56      TS23

Enter OLD CoilName to be changed (eg: HEAD) CHEST

Valid coils are as follow:
2_BREAST      ENDOREC      MRI QUAD WR   TRPA1234
8 TRPA test    EXTREM      NEUROV        NV MA
8CARDIAC      GPFLEX      NVARRA
8HRBRAIN      HEAD        NVHEAD
8HRBRAIN-DWI  HI RES BRAIN NVPA
8NVARRAY      HRBRN
8NVHEAD      HRBRN ONE RCVR P31_FLEX      US BILATBRAIN
AB FLEX MA    HRWRIST      PELVIC        US Cardiac Torso
AB FLEX MAPA  KNEEPA      POSTNECK      US KNEEQD
ANTNECK      LGEXTPA     PVLOWER       US LFTBRAIN
ATD3ENDO     LS456      PVMIDDLE      US NECKQD
ATDTORSO     LS56      PVUPPER       US NEUROPA
BODY         L_BREAST    QUAD T/L      US PROSTATE
BODYFLEX     Long Bone Lower QUADCSP      US RGTBRAIN
BREAST       Long Bone Upper QUADKNEE     US SHOULDER PA
BREAST MAPA  MAPVBOT     QWRIST MA     USCS12
BREASTPA     MAPVMID     R_BREAST      USCS123
BREASTPA Left MAPVTOP     SHLDRPA4     USCT234
BREASTPA Right MPFLEX MAPA  SHOPA2        USCTLBOT
CARDIAC      MRI AUTOTL  SHOPA3        USCTLMID
CARDIAC_ANT  MRI AUTOTL PA SHOULDER      USCTLTOP
CARDIAC_POST MRI BREAST  SHOULDERMAPA  USLS456
CS12         MRI BREAST TR SM GPFLEX MA  USLS56
CS123        MRI EXTRM LG SMEXTPA     USTL345
CT234        MRI EXTRM SM TL345        USTL45
CTLBOT       MRI PA EXTRM LG TL45        USTC23
CTLMID       MRI PA EXTRM SM TMJ
CTLTOP       MRI PA WRIST TMJP
DUAL         MRI PEDARRAY LG TORS
DUALFLEX     MRI PEDARRAY SM TORS
ECOILPA      MRI PEDHEAD LG TPLOWER
ENDOATD      MRI PEDHEAD SM TPUPPER
  
```

1. Change directory and start AllCoilCorrect.sh script.

2. Type the coil name to change and push the <Enter> key.

3. Type the coil's new name and push the <Enter> Key.

4. Type y <Enter> to confirm.

```

Enter NEW CoilName to be changed (eg: NEWHEADCOIL) TORSO
The old coil name "CHEST" will be changed to "TORSO" across
all protocols under SITE category.
Please Verify and Confirm [y/n] Y
Confirmed: change the coil name CHEST to TORSO
Changes made to... /usr/g/protocols/site/other/BOLUS
dc@b11b}[23]
  
```

UPDATING CUSTOMER PROTOCOLS WITH NEW COIL NAMES

ILLUSTRATION 9-2

5. Confirms the change was made. Repeat as necessary.

5. Repeat step 4 for all coils that may have renamed.

9-3 Having Coil Definition File Updated

This section covers path (3) on Illustration 9-1. If no information can be found for a coil then reference the **Coil Identification Guide** document from Support Central - MR on the GEMS Intranet as follows: **Manuals → Signa Documentation → Coils – 10.X and above. Signa Infinity & Excite Technology → Q.P. 11.x Coil Listing**. If the coil is not listed in this document then use the **Coil Report** form at the same location to notify the MR Coil Team.

They will research the coil and advise you as soon as possible of the new parameters. If the coil is found in the document or if you suspect the information in the Coil Definition file on the system is wrong, then a new Coil Definition file should be obtained. Two methods are available for updating a new Coil Definition file. The easiest and recommended method is to have the AutoSC download the file to sites that have an InSite connection. The next best method, especially for sites that do not have an InSite connection, is to obtain the latest listing of the CoilConfig.cfg.txt file for Release 10.0 from the same location as shown above on the Support Central – MR site. From this site you can view or copy the file to get the latest coil values. You will then need to use the Config File Manager tool to manually edit the coil name

Note

Do not Load a 10.x Coil Config File onto a 11.x Site

9-3-1 Downloading Latest Coil Config Definition File from AutoSC (Recommended)

This method is for sites with an InSite connection. The AutoSC request will download the following files to the /usr/local/bin directory for the site you requested:

- CoilConfig.cfg.1.5T (coil template file used by Restore Manager)
- CoilConfig.df.1.5T (coil template file used by Config File Manager tool)
- coildefs.sh (install script for template files)

Note: Type the following to determine the version (date) AutoSC coil template files at the site (example shown):

```
{sdc@t15}[1] cd /w/config <Enter>
{sdc@t15}[2] ls -l *autosc* <Enter>
-rw-r--r-- 1 sdc informix 133181 Sep 30 15:33 CoilConfig.cfg.1.5T.autosc
-rw-r--r-- 1 sdc informix 133268 Sep 30 15:33 CoilConfig.df.1.5T.autosc
```

9-3-1-1 Requesting Updated Coil Files from AutoSC

This process downloads new coil template files to the site, but does not change any site coil values (that's done in a later section).

1. Send an email request to the Auto Support Center (ASC) with the following information:
 - a. Address the email to **AutoSC (med)**
 - b. In the email "Subject:" heading, type: **sweep**
 - c. In the message body, enter the following lines:

```
type = coildefsll
system id = <systemid>
```

return mail = <email_address> (optional – Only if you want the return mail sent to a different address than sending).

d. Send the email. (Typical AutoSC response time is ~0.2-3 hours.)

2. **Download not successful:** If you receive an email that the download to the site not was successful, either troubleshoot the connectivity problem and try the request again, or proceed to Section 9-3-2, Quickplace Coil Update (Manual Method).

Download successful: You will receive an email that the download to the site was successful (see example). Continue with step 3.

```
From: autojob@iscmed.med.ge.com [mailto:autojob@iscmed.med.ge.com]
Sent: Tuesday, October 01, 2002 12:00 PM
To: Mruthik, Srikanth (MED)
Subject: AutoSC Notify, System <System ID>
```

DRIVER.SH was run. Please review for more information.

Please do not reply directly to this message. Questions and/or comments should be addressed to: asc_core@iscmed.med.ge.com

```
SYSTEM ID:      <System ID>
USN:            00000<System ID>
MLN:            9999
Site Name:      MR
Modality:       MR
Product Line:   LX
Date:          2002:10:01:12:05:42
```

- - - - -

```
EXCITE CoilUpdate files successfully downloaded
Downloaded at (MM.DD.YY_hh:mm:ss) : 10.01.2002_11:57:08
```

3. At the site, on the service desktop, click [**C-Shell**].
4. Type the following in the command window (do not login as “root”):

```
> cd /usr/local/bin <Enter>
> coildefs.sh <Enter> (This script backs up the original CoilConfig.cfg.1.5T and .df.1.5T
files, then installs the new ones in the /w/config directory. The existing site CoilConfig.cfg
file (contains sites coil names and values) is not modified with this script!). When the
script completes, the following is displayed:
```

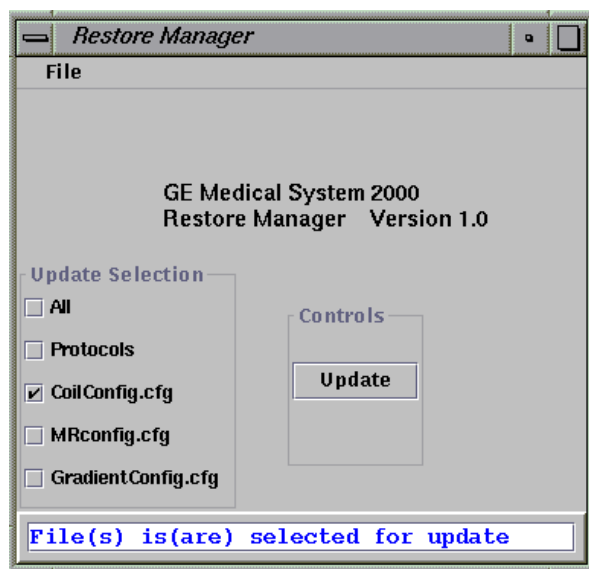
```
-----
Done with the CoilDefs update...
Please proceed with the next steps in the Service documentation ...
Exiting...
```


9-3-1-2 Using Restore Manager to Update All Coil Values (Automated Method)

This method first makes a backup copy of the site's CoilConfig.cfg file. It then reads each coil name from the CoilConfig.cfg (site's coil list) file and copies the coil parameters from the new downloaded CoilConfig.cfg.1.5T template file for each coil. If any coil name is not found in the template file, it is displayed in a popup message as an "unknown" coil.

1. Type the following in the command window (do not login as "root"):

runrestore <Enter> (Starts the Restore Manager GUI; see Illustration 9-3.)



RESTORE MANAGER

ILLUSTRATION 9-3

3. Select "**CoilConfig.cfg**" and then click **[Update]**. This copies the latest parameters for all site coils from the new CoilConfig.cfg.1.5T file to the CoilConfig.cfg (site's coil file).

Note

If any "unknown" coils are found, complete step 4 and then perform the steps in **Appendix G - Adding "Unknown" Coils Back In the Site's Coil Config File.**

4. Click **[Quit]** when the update is finished.
5. Reboot Signa for coil changes to take effect.
6. Perform a Head or Body scan to verify proper system operation.

9-3-2 Getting the Latest Coil Config Template from Quickplace (Manual Method)

This method is for sites without an InSite connection. The latest version CoilConfig.cfg file for Release 10.0 is available on the Coils – Excite Quickplace. From this site you can view or copy the file to get the latest coil values. You will then need to use the Config File Manager tool to manually edit the coil name

- CoilConfig.cfg.1.5T.txt (viewable version of coil template file; not a Unix file)
1. You can access the “Coils – Excite” Quickplace as follows: The latest copy is available from Support Central - MR on the GEMS Intranet as follows: **Manuals → Signa Documentation → Coils – 10.X and above. Signa Infinity & Excite Technology → Q.P. 9.0X and 10.X Coil Listing.**
 2. Scroll to the bottom of the Quickplace screen and select the “**CoilConfig.cfg.1.5T.txt**” document.
 3. Copy the file to your PC, or open the file and view/record the values for the coil(s) you need. (This is a viewable only file. It is not a Unix file and cannot be transferred to Signa.)

9-3-2-1 Using Config File Manager to Update Selected Coils

This method is used to change only those coils that need updating.

1. From the service desktop Utilities menu, select the **Configuration File Manager** tool.
2. Click **[Coil Config File]**. From the “Site Coils” list, select the coil names that need to be updated then click **[<<REMOVE<<]**. (Example: If NVHEAD and NVARRAY coils need be updated due to coil faults, select **NVHEAD** and **NVARRAY** then click **[<<REMOVE<<]**).
3. From the “Available Coils” list, select the same coil names as the last step (example: **NVHEAD, NVARRAY**), and then click **[>>ADD>>]**.
4. When all coils to be updated have been removed and re-added, exit and save changes by clicking **[Quit]**, **[Yes]**, and then **[Save]** to save changes.
5. Reboot Signa for coil changes to take effect.
6. Perform a test scan with the changed coil(s).

10- LOADING THE COILS TOOLS CD REV B SOFTWARE

A CDROM labeled “Coils Tools CD Rev B 2358042” is included in the software collector and should be loaded at this time. This CDROM must be loaded first so that current and later versions of the 8-channel High Resolution Brain Coil SNR software (this software is included on a separate CDROM and is shipped with each 8-channel High Resolution Brain Coil) can be installed and run. This must be loaded after the Service CDROM (Restricted or Advanced) has been loaded.

1. Place CD in host CDROM drive.
2. On the service desktop open a C Shell window.
3. Type **su** and press **<ENTER>**. When prompted for a password type **operator** and press **<ENTER>**.
4. Type **mount /mnt** and press **<ENTER>**.
5. Type **cd /usr/local/bin** and press **<ENTER>**.
6. Type **installCD** and press **<ENTER>**.
7. When the command prompt returns, type **exit** and press **<ENTER>**.
8. Remove the CD from the drive.

11- INSTALLING THE "PATCH_10_M4_SPECKEY" AUTHENTICATION CD KEY



THIS KEY MUST BE INSTALLED BEFORE THE SYSTEM WILL PERMIT CUSTOMER SCANNING. THE CUSTOMER WILL NOT BE ABLE TO RUN PATIENT SCANS ON A SYSTEM EQUIPPED WITH M4 SOFTWARE WITHOUT THIS KEY.

This key must be installed before the system will permit customer scanning. The customer will not be able to run patient scans on a system equipped with M4 software without this key. Service scanning and calibration can be done without the key. This key should be installed only after the Field Engineer has confirmed that all system calibrations have been completed and has verified that all system calibration and test results are meeting specifications. By installing this key the Field Engineer affirms that these actions have been completed. This CDROM should be shipped in the same box with the M4 10.0 software until June 2003. After June 2003 the key must be generated and then downloaded as a file or sent to the site on a CDROM. Please either call the OnLine Center or visit the Support Central - MR site if the CDROM is not included with the M4 10.0 software or is lost. This key is saved during the SaveINFO process and is restored when a RestoreINFO is performed. It should not be necessary to reload the key during the next software reload or load as long as a RestoreINFO is performed.

The System-in-Spec Authentication Key CD 2372933 should only be loaded on Signa 10.0 M4 software.

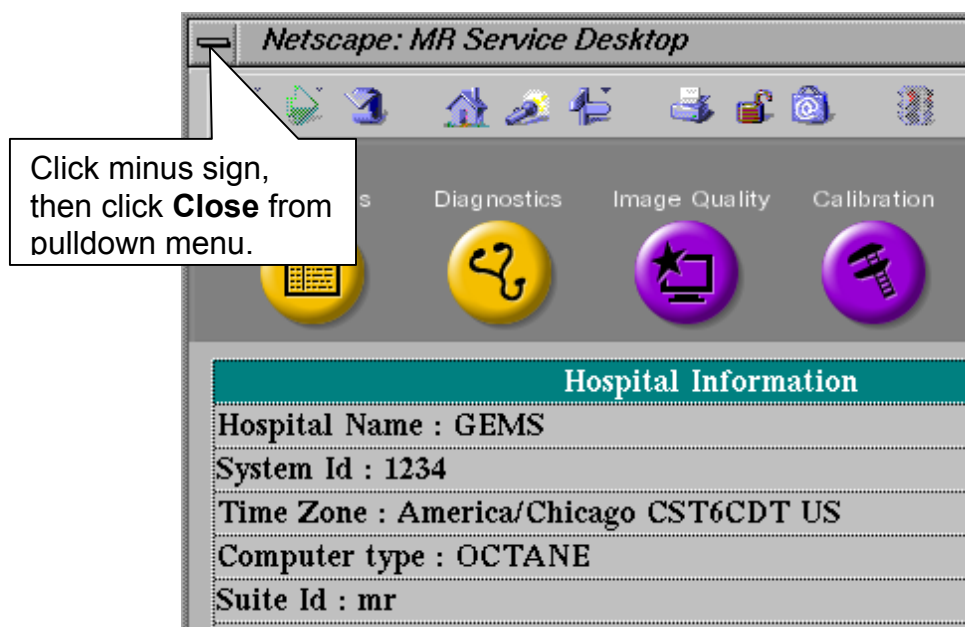
Special Instructions for loading this key:

This key can be loaded during LFC or after LFC. Reboot required after installation of key.

3- LOADING SIGNA 10.0 M4 SYSTEM-IN-SPEC KEY FROM CDROM

1. Confirm that the IRIX software is booted and that the applications software is running.
2. Place CD in host CDROM drive.
3. On the service desktop open a C Shell window.
4. Type **su** and press **<ENTER>**. When prompted for a password type **operator** and press **<ENTER>**.
5. Type **mount /mnt** and press **<ENTER>**.
6. Type **/usr/local/bin/installCD** and press **<ENTER>**.
7. When the command prompt returns (in about 15 seconds), type **exit** and press **<ENTER>** and then type **exit** and press **<ENTER>** again to close the C Shell window.

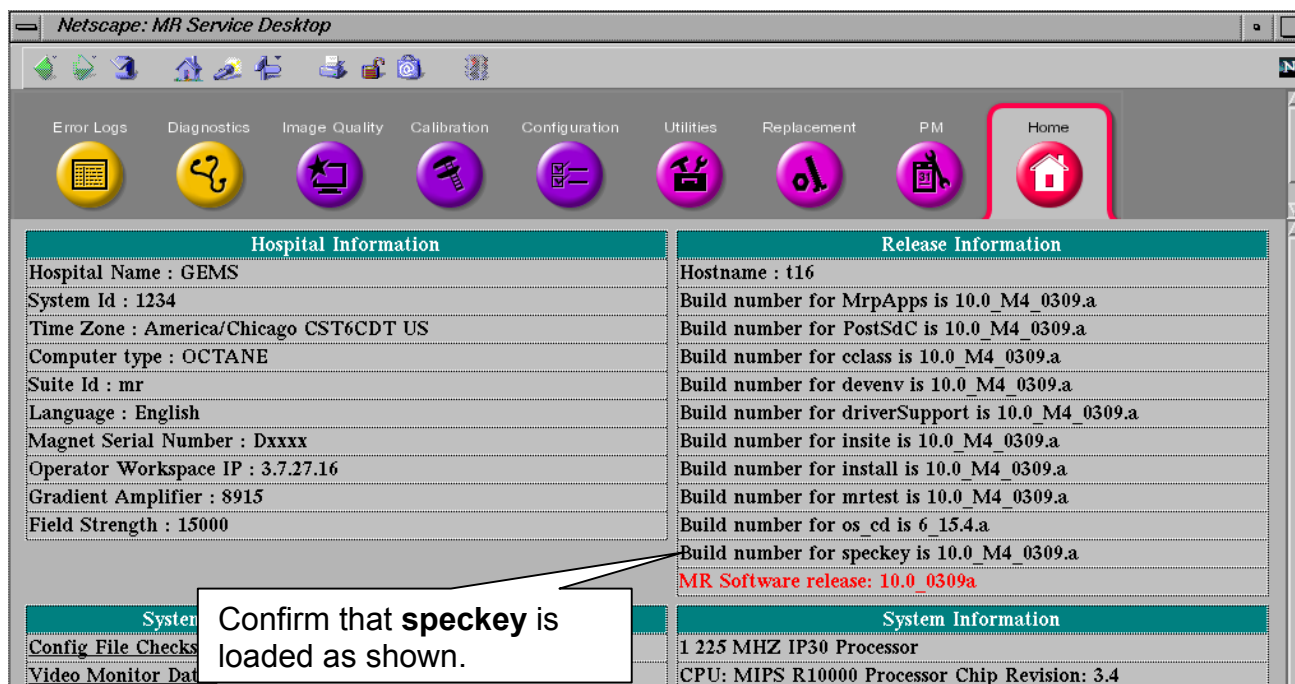
8. If not already in the ToolBelt Desktop then select the Toolbelt icon  to enter the ToolBelt Desktop.
9. If the MR Service Desktop (also referred to as the Common Service Desktop) is already running then close it by selecting the minus sign in the upper left corner of the window and then select **Close** from the pulldown menu. See Illustration 11-1.



CLOSING MR SERVICE DESKTOP
ILLUSTRATION 11-1

10. Click the **[Service Browser]** button on the Service Desktop Manager to restart the MR Service Desktop.

11. The MR Service Desktop should open in the Home page. Confirm that the key is listed in the first quadrant of the Home page titled "Release Information" as shown in Illustration 11-2.



CONFIRMING SPECKEY INSTALLATION
ILLUSTRATION 11-2

12. If the key is not listed then close and reopen the MR Service Desktop and check for it again. If it is still not listed then open a C-shell and repeat the installation process again.
13. Press the eject button on the front of the host CDROM drive and remove the CD from the drive.

APPENDIX A - HOST SOFTWARE INSTALL WORKSHEETS

Host IP Address: The Host IP should be obtained from the site network administrator if the scanner will be connected to the sites internal network. Or obtain an address by calling the OnLine Center (OLC) MR Support.

IWS PC IP Address: The IWS PC IP should be obtained from the site network administrator if the scanner will be connected to the sites internal network. Or obtain an address by calling the OnLine Center (OLC) MR Support.

FE Laptop IP Address: The Laptop IP should be obtained from the site network administrator if the scanner will be connected to the sites internal network. Or obtain an address by calling the OnLine Center (OLC) MR Support.

TPS Subnet Address: The TPS is on a subnet of the Signa LX scanner. A default address is loaded during the load from cold process and it should not be changed.

Hostname: This is the network name given to the scanner host computer.

Hospital Name: The Name of the hospital or other facility where the Horizon LX is installed. Name can include characters and/or integers up to 32 characters.

Doctors Title: The title to be used at the scanning site by the reading physician (Radiologist or Diagnostician).

Suite ID: Name of the Suite (if any) the scanner will be connected to. Leave blank otherwise.

Unique ID: The Unique System ID is a 16 alphanumeric character string assigned at the time the system is first installed. The Unique system ID is assigned by GE CARES when the first dispatch is opened to document the start of installation.

Service ID: The Service ID is a 16 alphanumeric character string assigned at the time the system is first installed. This is the same number that GE CARES uses to track financial and administrative data for the system.

Weight Unit: The weight system to be used by the scanner. Kilo Grams or US pounds.

Monitor type: The type of monitor the scanner will be using.

Windows 95 OEM Number: The Microsoft OEM Number found on the "Proof of Authenticity" for Windows 95. _____

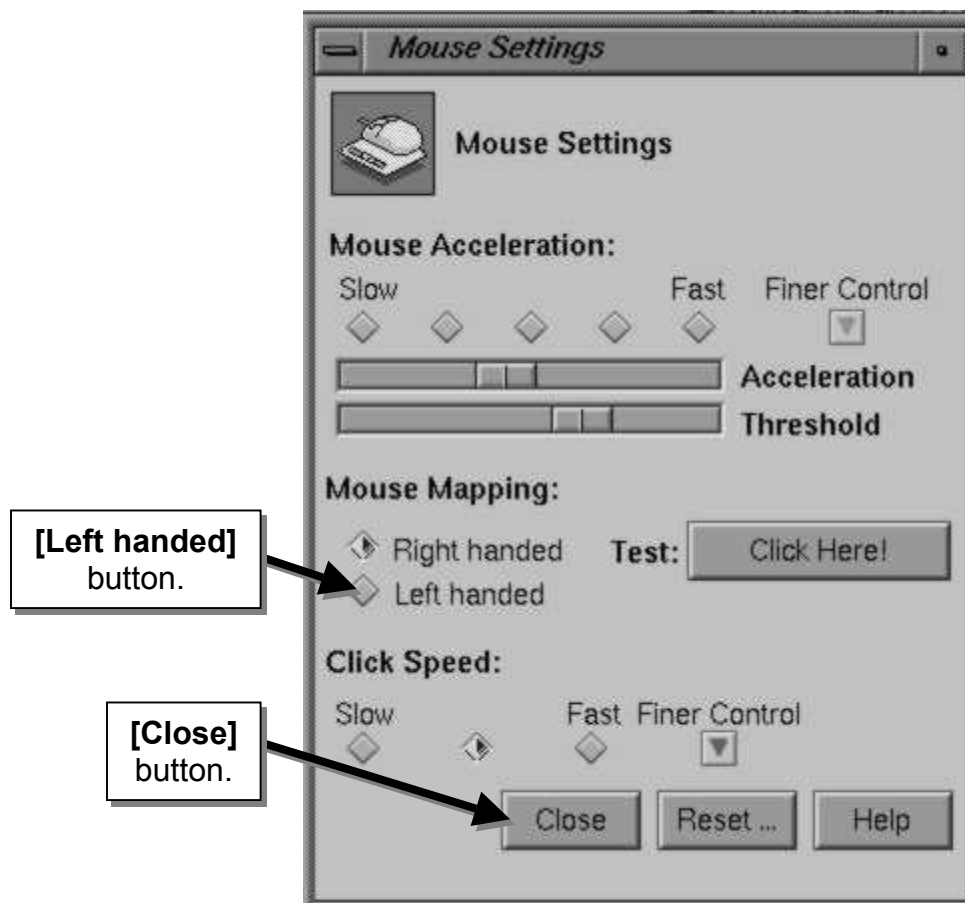
NetMask: For new installs, the Netmask should be obtained from the site network administrator if the scanner will be connected to the site's internal network. If this is a stand alone system, use the default netmask.

APPENDIX B - CONFIGURING MOUSE FOR LEFT HANDED USE

The system Host mouse is default configured for right handed use. It can be configured for permanent or temporary left handed use.

B-1 Configuring Mouse For Permanent Left Handed Use

1. If Signa is up and running, you must first log out and bring the system up to the Login screen. Otherwise, turn the system on and proceed to the Login screen.
2. Log in as **root**. The system default password is “**operator**”. On the root desktop, use the mouse cursor on the ToolChest and click and slide through **[Desk Top]** then **[Customize]** then **[Mouse]**.
3. Push the **[Left Handed]** button under Mouse Mapping, then push the **[Close]** Button to accept the changes. See Illustration B-1.



SET TIME/DATE MENU
ILLUSTRATION B-1

4. Log out of the root desktop and reboot the system. To change the mouse mapping back to Right handed. Repeat this section but push the **[Right Handed]** button, then reboot the system.

B-2 Configuring Mouse For Temporary Left Handed Use

Using this method will change the mouse mapping from right handed to left, but only until Signa is rebooted, when it will revert back to right handed mapping.

1. If not already at the Signa scanning level, go there. Open a command window and on the command line type: **mouse <Enter>**.
2. Push the **[Left Handed]** button under Mouse Mapping, then push the **[Close]** Button to accept the changes. (See Illustration B-1.)
3. To reconfigure the mouse back to right handed operation, repeat this section but push the **[Right Handed]** button, or reboot Signa.

APPENDIX C - DISK DRIVE LOW LEVEL FORMAT

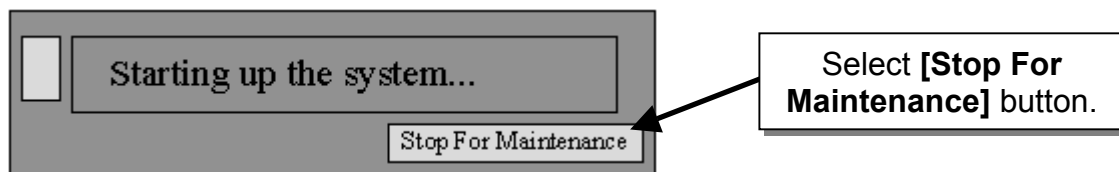
This section describes how to do a low level format of the disk drives. A low level format should only be done if loading a different Operating System (OS) or if suspected bad blocks on a disk drive are causing system problems. Depending on the system and disk drives, this procedure can last up to 1 hour per drive or as little as 20 minutes per drive.

1. Insert the **IRIX 6.5** CD labeled "**Boot/Update**" into the CDROM Drive on the workstation (right side).

Note

User responses are bolded text

2. With the system up and running, go to the service desktop and select **[Shutdown]**. If the system is not up and running, switch on power at this time.
3. When prompted, click on the **[Restart Button]**. Stop the reboot process by clicking on the **[Stop For Maintenance]** button when it appears.



4. Next, click on the **[Enter Command Monitor]** button when it becomes available. Type one the following:

```
>> boot -f dksc(1,3,8)sash64 dksc(1,3,7)stand/fx.64 --x <Enter>
```

5. Continue typing as follows:

```
fx: "device-name" = (dksc) <Enter>
```

```
.
```

```
fx: ctrlr# = (0) <Enter>
```

```
.
```

```
fx: drive# = (1) <Enter> or 2 <Enter> if doing the second drive.
```

```
.
```

```
fx: d/fo <Enter>
```

```
.
```

```
Press <Enter> to accept current parameters.
```

```
.
```

```
Type: Yes <Enter> to start the format.
```

6. The format can take up to 1 hour, depending on the system and disk drives that are being formatted. There will be no prompts to the user while the format is running. Be patient and do not interfere with the system. When the format is finished the system will prompt with:

```
Format completed successfully
```

```
.
```

```
.
```

```
fx/debug>
```

7. Reset or power cycle the computer to exit debug and proceed to Section 2 of this document to proceed with the OS (Operating System) load for the system.

APPENDIX D - GENERAL INFORMATION AND NICE TO KNOWS

D-1 Bore Temperature Sensor - Must Be Working!

If the bore temperature sensor is disabled, sites with iDrive software will have problems. Sites running the iDrive software must have a functional bore temperature sense circuit. The symptom for a non-functional Bore sensor will be evident when an iDrive scan is attempted; the prescan will complete OK, but the scan will stop immediately after the first pulse is heard in the body coil. No images will be produced and no errors will appear in the log.

If the system has iDrive software (as listed in *Option Key* tab of Guided Install), to verify functional bore temperature circuit, set up the scan in Table D-1 (or any other Real Time scan). Download the scan and select **[Scan]**. If prescan completes and the scan continues to run, you have a functional bore temperature circuit. If prescan completes OK, but the scan stops immediately after the first pulse, check the bore temperature circuit.

TABLE D-1
IDRIVE SAMPLE PROTOCOL

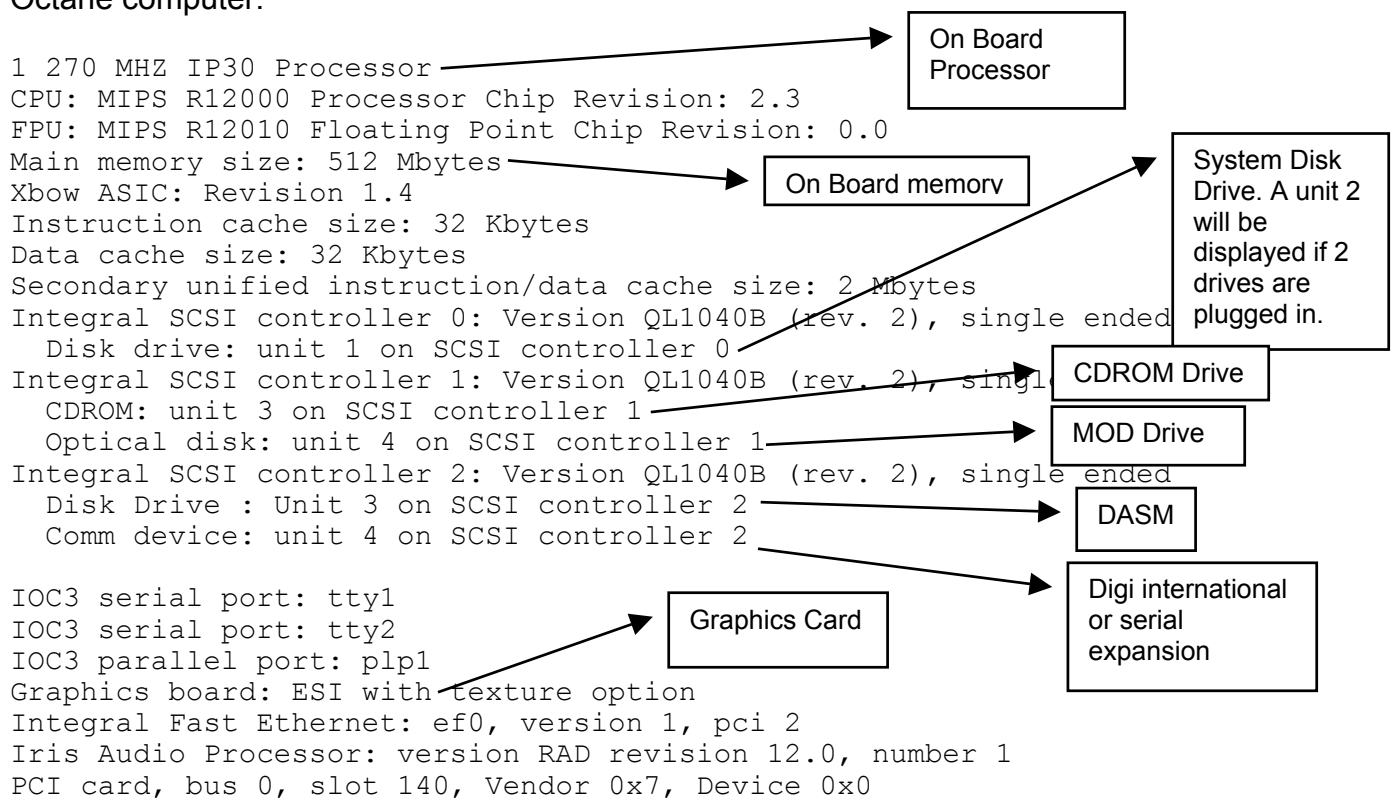
<u>PATIENT REGISTER</u>	[New Pt]	<u>SCANNING RANGE</u>	
<u>PATIENT INFORMATION</u>		FOV.	[32]
Patient Id	geservice	Slice Thickness.	[7]
Patient Name	test	Spacing	[1.5]
Weight (Lb)	111	Start	0
	[Landmark]	End..	0
Landmark	[Sternal Notch]	# Slices	1 (default)
		L/R Center	0 (default)
<u>PATIENT PROTOCOLS</u>	[Patient Position]	P/A Center	0 (default)
		Table Delta	0.00 (default)
<u>PATIENT POSITION</u>		<u>ACQUISITION TIMING</u>	
Patient Position	[>] [Supine]	Freq	[128]
Patient Entry.	[>] [Head First]	Phase	[128]
Coil	[...] [Body] [Accept]	NEX	[1]
<u>IMAGING PARAMETERS</u>		Phase FOV	1.00 (default)
Plane	[>] [Axial]	Freq Dir	[>] [R/L]
Mode	[>] [2D]	Auto Center Freq	[>] [water]
Pulse Seq	[...] [Fast GRE] [Accept]	Autoshim	[on]
Imaging Options	[Real-time]	(lowest window)	[Save Series]
Psd Name	no entry		
Protocol	no entry		
<u>SCAN TIMING</u>			
# of Echoes	1 (default)		
TE	[Min Full]		
TR.	[Minimum]		
Flip Angle	[20]		
Bandwidth	[62.50]		

D-2 Signa Power up Delay

When Signa is first powered on there is a considerable period of time that nothing is seen on the monitor; up to 4 minutes depending on the computer type. The system is checking configurations and starting software processes at this time. If you are concerned that the system is hung, push the **<Esc>** button on the top left side of the keyboard. The system will then post to the monitor all its boot up activities.

D- 3 [hinv] Command

The **[hinv]** command can be used to check the configuration of the operator workspace. To use the command, open a command window and at the prompt type: **hinv <Enter>**. The following is an example of an **[hinv]** output of a common configured Signa system using Octane computer.



APPENDIX E - SERVICE SOFTWARE LOAD

E-1 Loading Service Software

1. Go to the install GUI window, and select the **ServiceSW/Picture this/InSite** tab.
2. Proceed to Section E-2, Load Advanced Service Software, **or** Section E-3, Load Restricted Service Software (for sites with GE Service Only).

(Restricted Service Software is only available for GE use; Advanced Service Software is only available to sites with a valid Advanced Service Package Limited License.)

E-2 Load Advanced Service Software

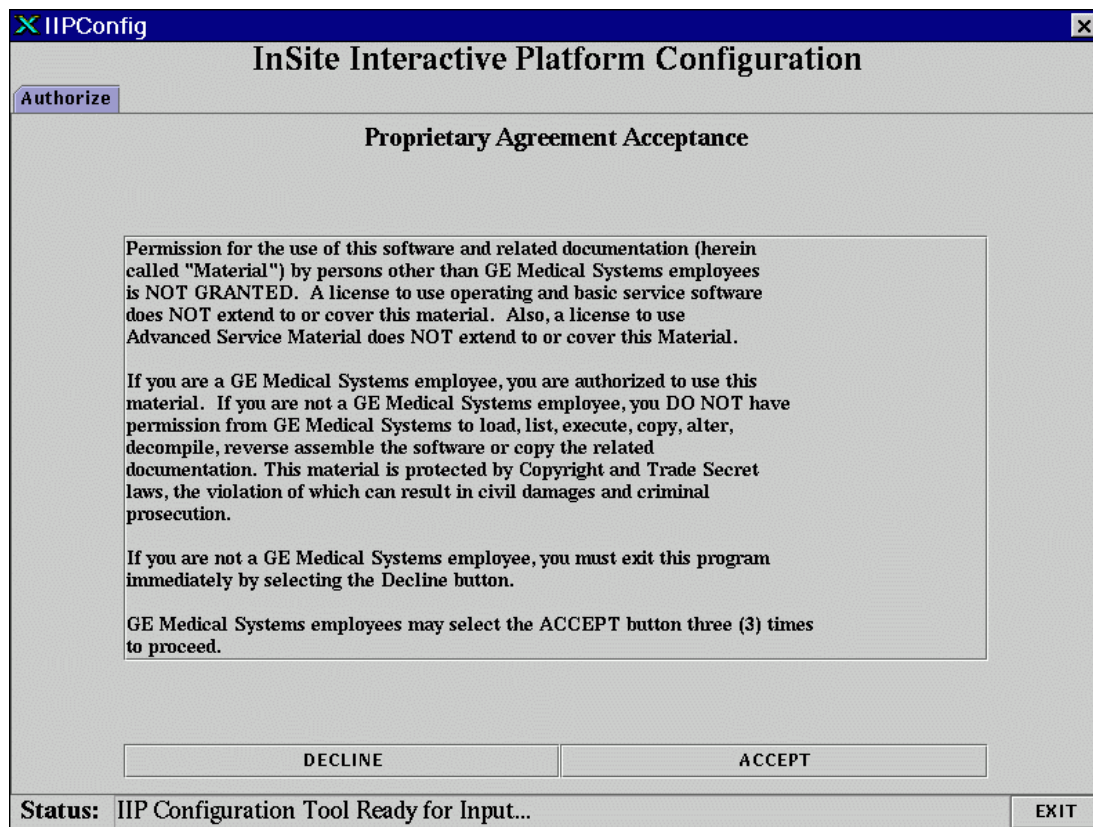
To load Advanced Service Software on the system:

1. Place the Advanced Service CD-ROM 2167558, in the host CD-ROM drive.
2. Select **[Load Advanced Service]**.
3. Read the proprietary message window, and respond accordingly.
4. A message "Advance software load in progress..." will display. The load will take about 5 minutes.
4. Wait for the message "Advanced Service load done" in the Install GUI status bar (at bottom of window).
5. Proceed to Section 6-5, Option Keys and Verification of Restored Information.

E-3 Load Restricted Service (for sites with GE Service only)

To load Restricted Service Software on the system:

1. Place Restricted Service CD-ROM 2167557, in the host CD-ROM drive.
2. Select **[Load Restricted Service]**. Read the proprietary message and respond accordingly.
3. A message "Restricted software load in progress..." will display. The load will take about 14 minutes. Be patient. Illustration E-1 will display near the end of the Restricted software load.



PROPRIETARY AUTHORIZATION TAB

ILLUSTRATION E-1

4. GEMS Field Engineers loading GEMS Restricted Service Software should configure InSite at this time. Select **[Accept]** 3 times and proceed to Section E-4, Configuration for Prodiags and InSite Checkout.

If you do not have an InSite modem or you will configure it later, select **[Exit]**, and proceed to Section E-5, Option Keys and Verification of Restored Information.

E-4 Configuration for ProDiags and InSite Checkout (for sites with GE Service)

The instructions in this section will step you through the minimum amount of screens to enable the basic functionality needed to enable ProDiags and complete an InSite Checkout with the OnLine Center. For information on the CUSTOM setup in the InSite Configuration GUI, refer to the procedure "Insite/IIP Installation Signa 8.3 or above" on the Signa 8.x Service Methods CD-ROM (2160623-1).

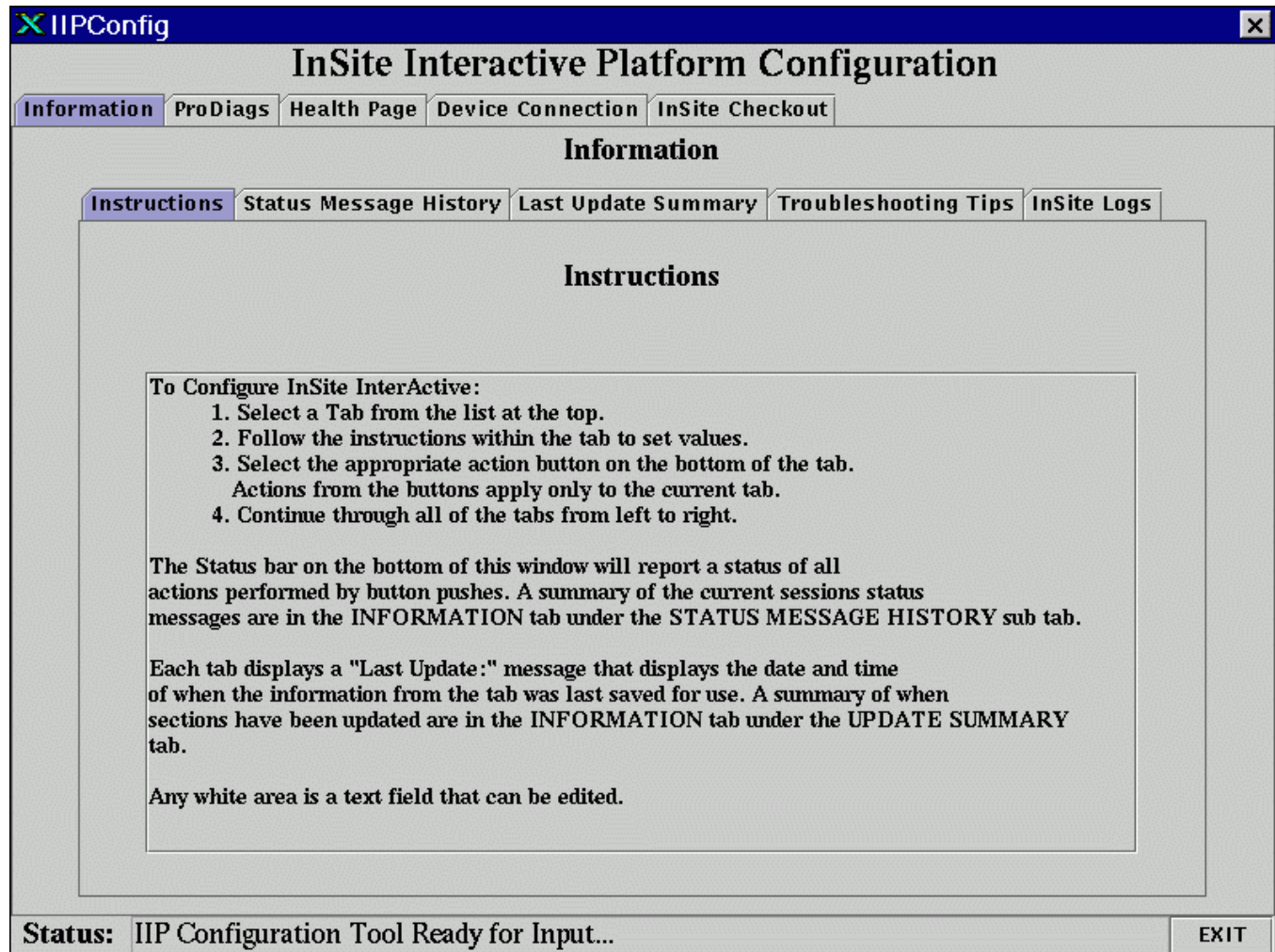
Note

Pro Diags are defaulted to call out to the ASC daily. If the attempt to call out fails, the software reattempts to call out hourly until successful. If the modem or network connection is down, **these hourly attempts to call out can slow down the scanner**. It is suggested that during InSite IIP setup, ProDiags call out to ASC be configured to dial out weekly and reattempt daily

Note

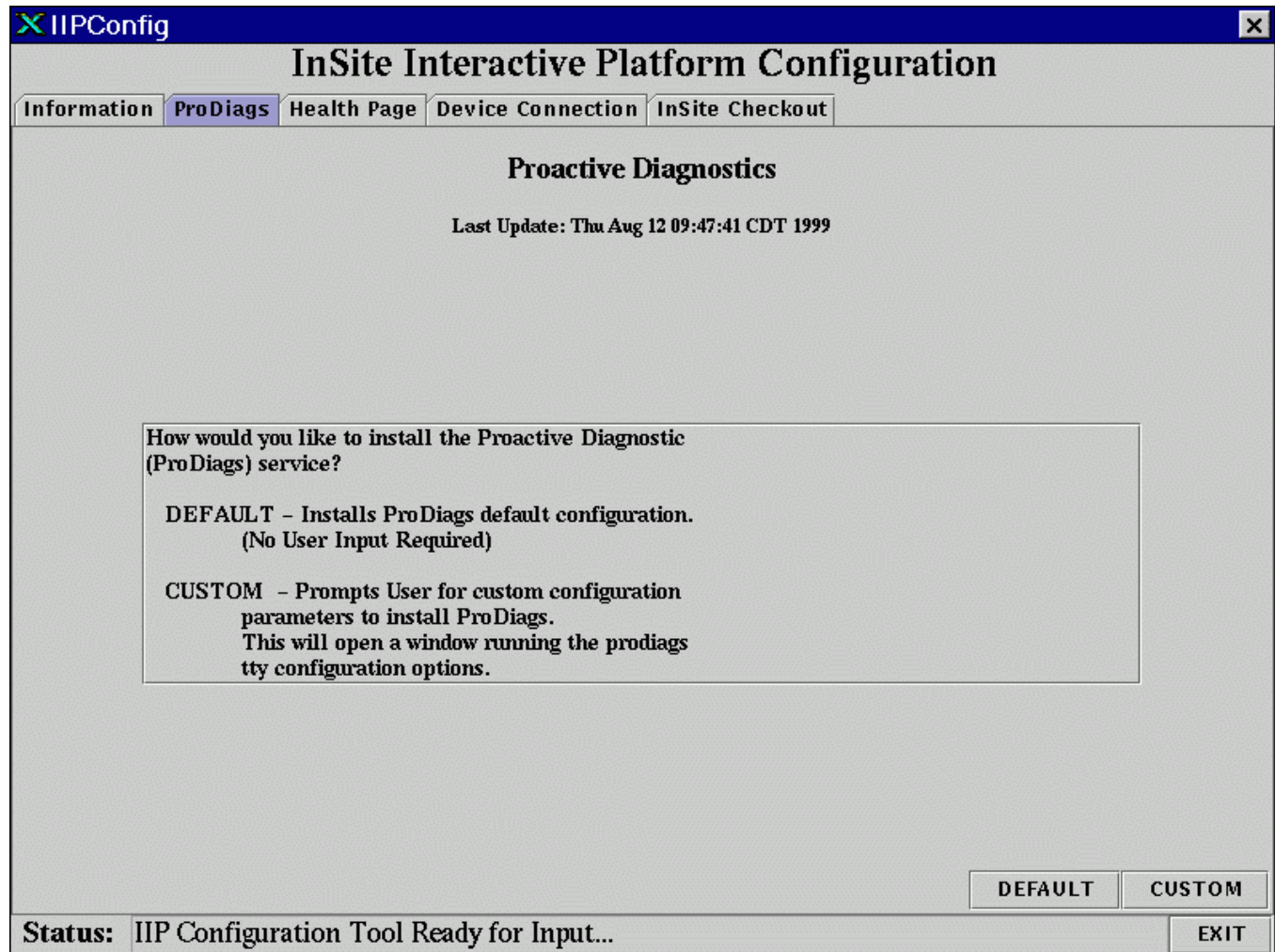
Due to software changes at the OLC to support iLinq (formerly called InSite Interactive), all 8.3 systems must initially do a Manual Checkout (i.e **FE must call the OLC for initial checkout**). Once this has been done, the Auto Checkout feature can be used for future reloads.

1. Following acceptance, the main tabs are initialized. See Illustration E-2.



INFORMATION TAB, INSTRUCTIONS SUB-TAB
ILLUSTRATION E-2

2. Select **ProDiags** tab to display screen as shown in Illustration E-3.
3. Select **[Default]**. This will configure the ProDiags feature with default schedules provided with the software. The Health Page screen will then be displayed. See Illustration 6-4. A status message "ProDiags Auto Setup: Completed Successfully" should be displayed at the bottom of the screen.



PROACTIVE DIAGNOSTICS TAB
ILLUSTRATION E-3

IIPConfig

InSite Interactive Platform Configuration

Information ProDiags **Health Page** Device Connection InSite Checkout

Health Page

Last Update: Wed Aug 18 09:56:36 CDT 1999

Enter or edit E-mail addresses to receive the Health Page report.
You MUST enter at least one address.

NOTE: Include the full Internet address or you will not receive the report!
Example: FirstName.LastName@med.ge.com

DEFAULT - Updates the Health Page E-mail addresses only.

CUSTOM - To gain access to Custom Health Page settings.
NOTE: E-mail addresses MUST be entered before
entering Custom Health Page

☐ HealthPage Report Disabled

Enter Address List

DEFAULT CUSTOM >>

Status: IIP Configuration Tool Ready for Input... EXIT

HEALTH PAGE TAB
ILLUSTRATION E-4

- Click in the **Enter Address List** box and type in a valid e-mail address. A valid e-mail address is defined as a string in the format **<name> @ <location>**

The format for GEMS e-mail addresses is: *<firstname> . <lastname> @med.ge.com*

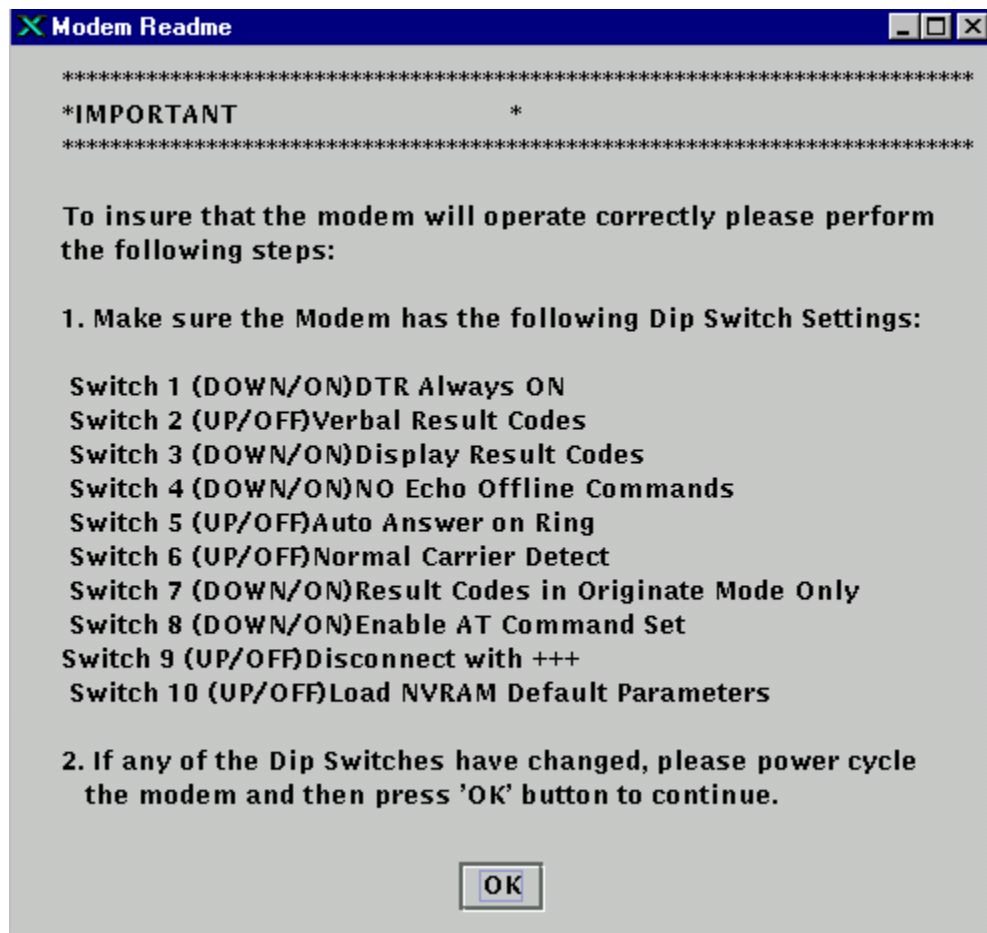
Be careful to enter correct e-mail address(es). All e-mail addresses are saved to the */usr/g/config/healthpg.adr* file in the format: *user<#>=<email address>*.

- Select the **[DEFAULT]** button to configure Health Page with the default schedule file. The Device Connection screen will then be displayed. See Illustration E-5. A status message "Health Page Email... Completed Successfully" should be displayed.

The screenshot shows a window titled "IIPConfig" with a sub-header "InSite Interactive Platform Configuration". Below this is a navigation bar with tabs: "Information", "ProDiags", "Health Page", "Device Connection" (which is selected), and "InSite Checkout". The main content area is titled "Connection Device Configuration" and shows a "Last Update: Fri Aug 13 15:50:42 CDT 1999". A "Device Connection Type" dropdown menu is set to "Modem". A text box contains instructions: "Select the InSite Connection Type above, or select the Modem settings given below and Press APPLY - To update the modem and modem configuration files with model and country defaults. CUSTOM - For other modem settings." Below this are several configuration fields: "Dial-out Prefix" (set to "8,"), "Dialing mode" (set to "Tone"), "Modem Type" (set to "USRobotics Courier V.34 or V.Everything"), "Country" (set to "Default - All Others"), "CPU Serial Port Name" (set to "/dev/ttyf1"), and "CPU Serial Port Speed" (set to "115200"). At the bottom right are "APPLY" and "CUSTOM >>" buttons. A status bar at the bottom left says "Status: IIP Configuration Tool Ready for Input..." and a button labeled "EXIT" is on the right.

DEVICE CONNECTION TAB
ILLUSTRATION E-5

6. Select a Dial-out prefix or type in a site specific prefix. Dial-out prefixes may be required on some sites to get a line outside of the hospital. " 9,1 " is the most common and is the default.
7. Select Modem Type that you have at this site. Note: Modem Type was recorded in Table 2-1.
8. Select **[APPLY]** to setup registers on the modem, and store values to the modem's NVRAM. When the **[APPLY]** button is selected, another screen will be displayed for the selected modem type (see example in Illustration E-6).

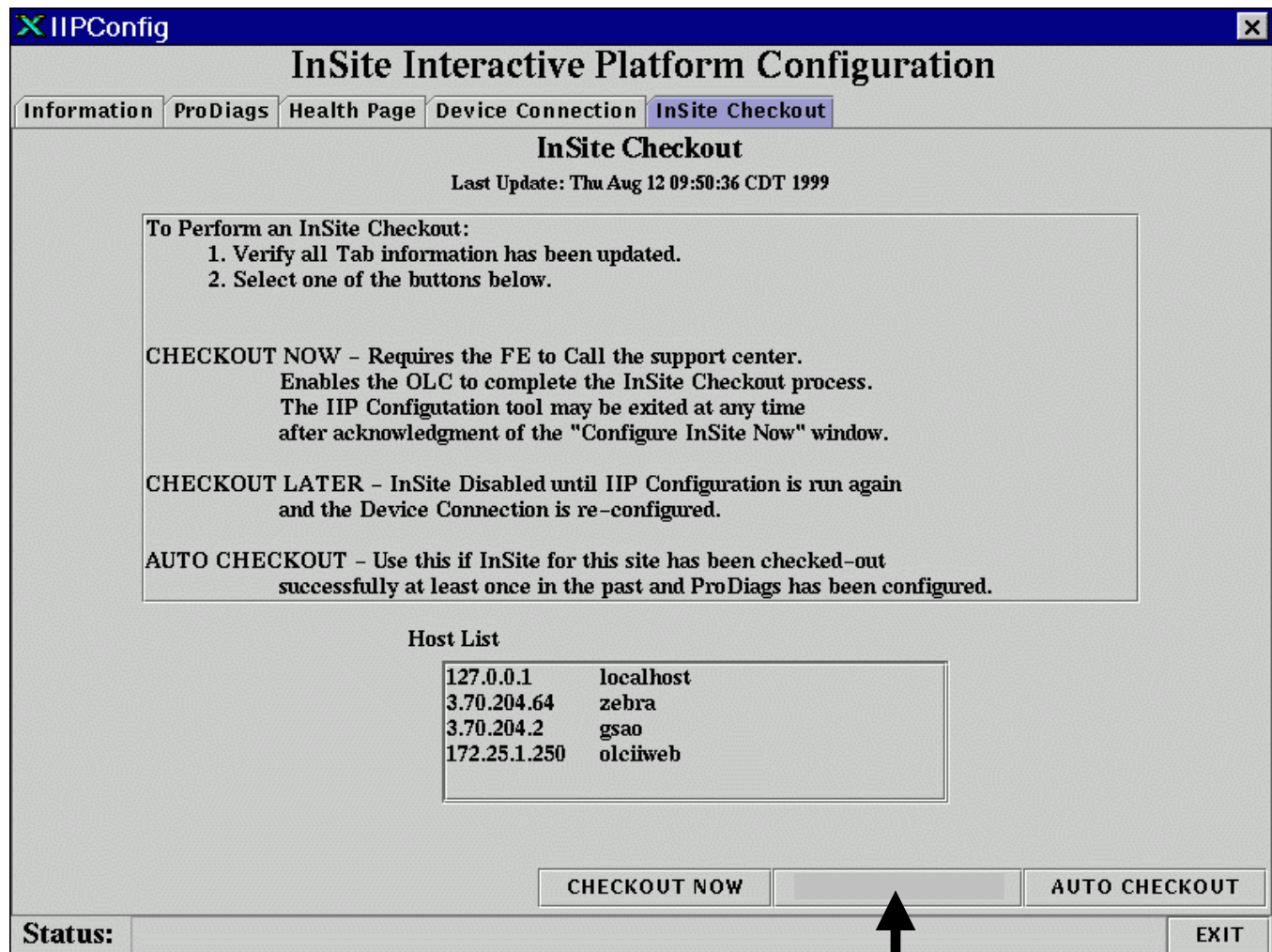


MODEM "README" SCREEN (EXAMPLE IS FOR USR V.34 SELECTION)
ILLUSTRATION E-6

NOTE

Be aware the InSite Configuration may recommend different switch settings for your modem as compared to the settings that worked with 8.2.5 software. Verify the modem switch settings match those recommended in the InSite Configuration.

9. After properly verifying/configuring any switches or other modem parameters on the modem, select **[OK]** which will do the programming of the modem. The InSite Checkout screen will then be displayed. See Illustration E-7.



INSITE CHECKOUT TAB
ILLUSTRATION E-7

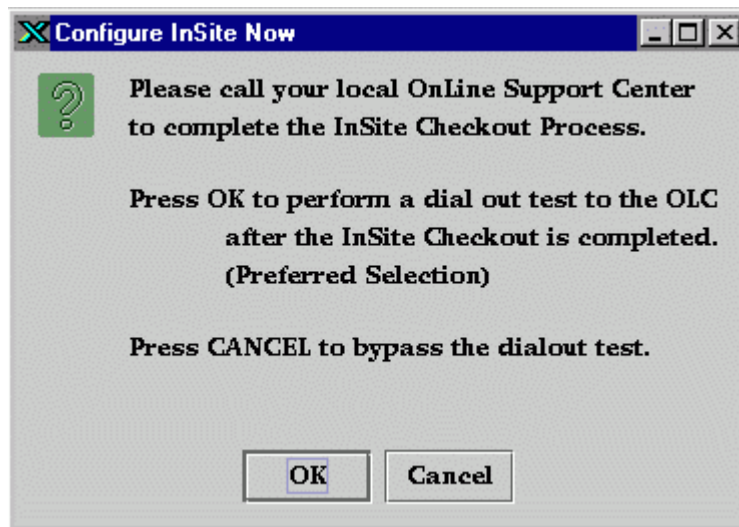
Disable Insite (ASP2 Software)

Checkout Later (8.3 Software or later)

10. The InSite Checkout Tab allows the user to complete the InSite configuration process. The three checkout options are described below, but you must select [CHECKOUT NOW].

- a) **[CHECKOUT NOW]** configures PPP to allow a dial in connection from the OnLine Center. The user must call the local OnLine Center to complete the checkout. See Illustration E-8.

If an InSite Checkout cannot be completed at this time, after selecting **[CHECKOUT NOW]**, select **[Cancel]** followed by **[Exit]**. This will bring up the Exit Confirmation Window (Illustration E-9). Select **[OK]** to exit the InSite Configuration GUI. Exiting in this manner will leave the serial port of the computer enabled allowing your OnLine Center to complete the checkout at a future time without requiring an FE be onsite.



CONFIGURE INSITE NOW REQUESTER
ILLUSTRATION E-8

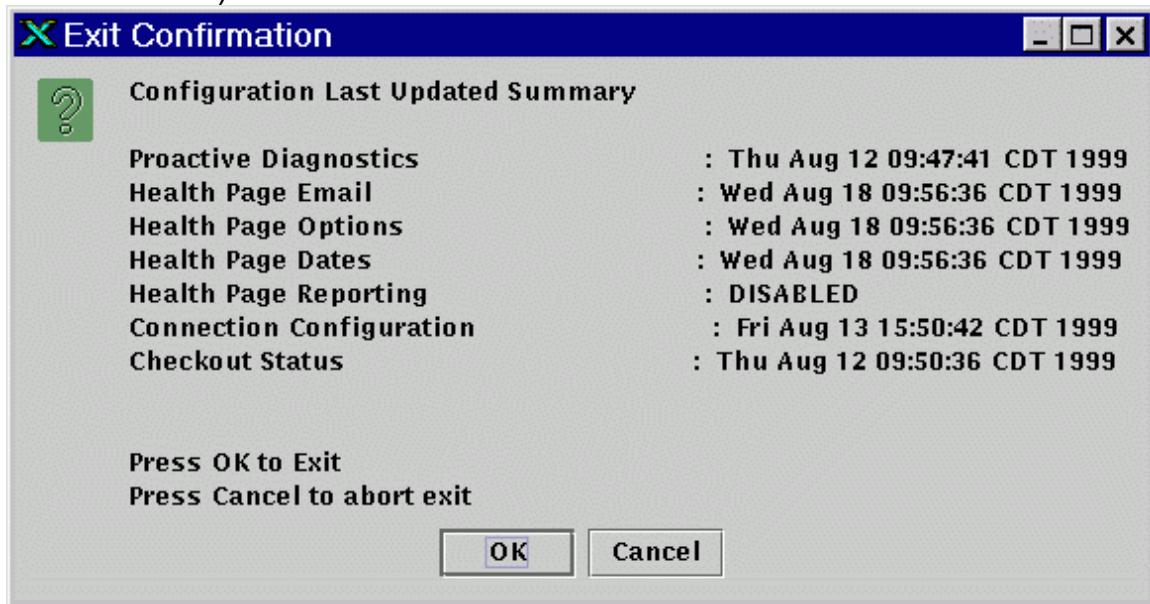
Note

When the user selects the **[CHECKOUT NOW]** button, the “Configure InSite Now” window is displayed. This provides the user the option to run a dial-out test upon completion of the InSite Checkout Process. If the user selects the **[OK]** button before the InSite Checkout Process is complete, the IIP Configuration Tool watches for a `~insite/sclink.cfg` file to be downloaded over the connection. The IIP Configuration Tool will only wait 10 minutes for the `sclink.cfg` file to arrive. When the file appears, the dial-out test starts. This test ensures the dialout prefix provided by the user and the number of the OnLine Center are correct. This also confirms that Proactive Diagnostics messages and **[Contact GE]** requests can be sent to the OnLine Center for sites entitled for Insite Interactive.

When the dialout test is in the process of connecting to the AutoSC, the IIP Configuration GUI will not update and may appear locked up until the connection to the AutoSC is completed or the connection times out. Please be patient.

- b) **[CHECKOUT LATER]** (8.3 and ASP1Software), **[Disable Insite]** (ASP2 Software). **Never select this** because it disables PPP and will require an FE to be on-site to invoke this screen for the OnLine Center to be able to complete an InSite Checkout. If this is selected by mistake, the user MUST rerun the IIP Configuration Tool, select the **[APPLY]** button on the Modem Tab, and then select the **[CHECKOUT NOW]** button to enable PPP for dial in.
- c) **[AUTO CHECKOUT]** is not used for this FMI because a manual checkout is required to be updated to the latest model type in the OLC database. The RestoreINFO process does restore the `/usr/g/insite/sclink.cfg` file needed for Auto Checkout to work. This button will only be active if a `sclink.cfg` file exists in the directory above.
[AUTO CHECKOUT] will run ConfigLink, dial the AutoSC, queue a task with the AutoSC to dial back, and wait for the AutoSC to dial back and leave a file `/tmp/autocheckout`. The IIP Configuration Tool will wait only 10 minutes for the file to appear. Typically, the AutoSC will call back in 5-6 minutes.

11. After selecting **[Exit]**, an updated summary is provided. See Illustration E-9. A valid date/time should be displayed for each function that you successfully configured. If any function shows "**NO FILE**", that function was not properly configured.
12. Select the **[OK]** button to exit IIP Configuration Tool.
13. Wait for the message "Restricted Service load done" in the Install GUI status bar (at bottom of window).



EXIT CONFIRMATION REQUESTER
ILLUSTRATION E-9

Important! For additional information about setting up or troubleshooting the Insite software/hardware, refer to the procedure "Insite/IIP Installation Signa 8.3 or above" on the Signa 8.x Service Methods CD-ROM (2160623-1).

APPENDIX F SAVEINFO BACKUP FILE AND LOG FILE

F-1 SaveInfo Backup File

Beginning with software release Signa 9.1 and 10.0, whenever a SaveInfo is made a backup file is created of all the files saved to the SaveInfo MOD. This backup file is located in **/usr/g/service/log/saveINFO.tar.gz** and can be used to manually update configuration files in the advent the SaveInfo MOD becomes corrupted. It is suggested that a backup copy be kept by using FTP to move the file to the service laptop or other machine and saved.

To use the contents of this file is a manual process at the moment. Following is some quick steps for file example file recovery:

1. Create a directory to move the backup file to. It is suggested to create the directory in **/temp**. If created there the files will be deleted on reboot saving disk space.

```
> cd /temp <Enter> (Move to temp directory)
> mkdir junk <Enter> (Create a directory named "junk")
```

2. Move or FTP your saved copy of **saveINFO.tar.gz** to **/temp/junk**
3. Now unzip and tar the SaveInfo file.

```
> gunzip saveINFO.tar.gz <Enter>
> tar -xvf saveINFO.tar <Enter>
```

4. You will now find a subdirectory tree created under **/temp/junk**. cd to **/temp/junk/usr/g/MRINFO/holdingtank** and you will find in that directory a directory tree similar to the Signa product directory tree. All the configuration files that were saved during the saveInfo can be found in the directory it came from on the system.

As an example assume the **MRconfig.cfg** file on your system became corrupted and you wanted to just replace that file. Unzip and tar the saveInfo file as shown above. Then do the following command line step to replace the file:

```
> cp /temp/junk/usr/g/MRINFO/holdingtank/usr/gvx/w/config/MRconfig.cfg
usr/g/MRINFO/holdingtank/usr/gvx/w/config/MRconfig.cfg <Enter>
```

In this example there is not enough room here to show it, but this whole statement is on one line with a space between **MRconfig.cfg** and **usr**. The system will ask if you want to overwrite the existing file. Answer yes.

Or, perhaps you may not want to view the contents of the older file and compare it's contents to what is seen in the current config file. Then you would only need to do the following:

```
> more /temp/junk/usr/g/MRINFO/holdingtank/usr/gvx/w/config/MRconfig.cfg
<Enter>
```


F-2 SaveInfo Log File

Beginning with software release Signa 9.1 and 10.0, SaveINFO now creates a log file in /usr/g/service/log directory called saveINFO.log. This file will list all the files that saveINFO was expecting to save but could not find on the system and hence could not save. Some of the files may be specific to TWIN or HFO or MGD systems and in those cases the log file will have a comment to that effect so that the FE is not confused. It is not required but is a good practice that this file is checked after a saveInfo to insure all files have been successfully saved.

APPENDIX G – ADDING “UNKNOWN” COILS BACK INTO THE SITE’S COIL CONFIG FILE

The section explains how to manually added “unknown” coils found by Restore Manager (Section 9-3-1-2) back into the site’s Coil Config file so they are available for customers to use.

1. Type the following in the command window:

```
> cd /usr/g/service/log <Enter>  
> more UnknownCoilList.txt <Enter> (Displays a list of “unknown” coils.)  
> more UnknownCoilDefs.log <Enter> (Displays the original coil parameters for each  
“unknown” coil.)
```

2. From the service desktop Utilities menu, select the **Configuration File Manager** tool.
3. Click **[Coil Config File]**, then click **[Add new Coil]**.
4. The *Add a New Coil* screen displays coil value fields for a new coil. Either click in the fields or tab through the fields to enter values from the “UnknownCoilDefs.log”. (Use the **<Delete>** key to edit text; the **<Backspace>** key works from the end of the field instead of where the cursor is located). Click **[OK]** to close the screen.
5. Repeat step 4 for additional “unknown” coils. When all “unknown” coils have been added, exit and save changes by clicking **[Quit]**, **[Yes]**, and then **[Save]** to save changes.
6. Reboot Signa for coil changes to take effect.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	Jan 07, 2003	Thome	Initial release for Signa 10.x
1	Feb 15, 2003	Thome	Added Section 10, made corrections to section 9.
2	Feb 17, 2003	Thome	Added Section 11.
3	Feb 19, 2003	Thome	Re-wrote Section 11 for installing key.
4	Feb 24, 2003	Thome	Re-wrote Section 11 for installing key.
5	March 3, 2003	Thome	Re-wrote Section 11 for installing key.
6	July 11, 2003	D. Hofstetter	Changed the title to section 5 to solve a cqa, added comments concerning SCSI Expansion box upgrade.